

Emerging Low-Dimensional Materials for Nonlinear Optics and Ultrafast Photonics

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Low-dimensional (LD) materials demonstrate intriguing optical properties, which lead to applications in diverse fields, such as photonics, biomedicine and energy. Due to modulation of electronic structure by the reduced structural dimensionality, LD versions of metal, semiconductor and topological insulators (TIs) at the same time bear distinct nonlinear optical (NLO) properties as compared with their bulk counterparts. Their interaction with short pulse laser excitation exhibits a strong nonlinear character manifested by NLO absorption, giving rise to optical limiting or saturated absorption associated with excited state absorption and Pauli blocking in different materials. In particular, the saturable absorption of these emerging LD materials including two-dimensional semiconductors as well as colloidal TI nanoparticles has recently been utilized for Q-switching and mode-locking ultra-short pulse generation across the visible, near infrared and middle infrared wavelength regions. Beside the large operation bandwidth, these ultrafast photonics applications are especially benefit from the high recovery rate as well as the facile processibility of these LD materials. The prominent NLO response of these LD materials have also provided new avenues for the development of novel NLO and photonics devices for all-optical control as well as optical circuits beyond ultrafast lasers.

1. Introduction

Structural low dimensionality plays an essential role in defining the physical as well as chemical properties of most nanoscale systems. Owing to the confinement effect arising from small size, low-dimensional (LD) materials offer intriguing properties that are not found in their bulk counterparts and have been consistently explored for applications ranging from photonics to bio-medicine. In semiconductors it is known as the quantum

confinement effect in which reduction in size results in spectral blue shift, while for noble metal nanoscale size can lead to localized surface plasmon resonance (LSPR) arising from collective oscillation of conduction electrons in response to light excitation. These effects manifest their strong size-dependent optical properties, such as absorption and emission. In particular, along with the dramatic change in linear optical response, the confinement of electron motion in nanoscale materials below a critical size at the same time strengthens their interaction with intense optical fields, leading to much enhanced nonlinear optical (NLO) response that involves completely new physical mechanisms. For instance, due to strong optical nonlinearity, semiconductor and metal nanomaterials have demonstrated attractive optical limiting (OL) and saturable absorption which are absent or far superior to their bulk counterparts. The NLO properties of nanoscale semiconductors, such as quantum dots, quantum wires and quantum wells, and noble metal nano-

structures have enabled diverse applications including optical switches and ultrafast laser pulse generation.^[1–4]

On the other hand, the milestone discoveries in nanoscience, such as fullerene, carbon nanotube (CNT) and graphene, have triggered an explosion of research into their physical-chemical characteristics and the exploiting of their potential applications. Beside the surprising structural diversity on the nanoscale, these carbon based materials show attractive transport, optical and mechanical properties distinct from their bulk counterpart graphite, which is built up with the same sp² hybridized C–C bonding. These unique structural features combined with the peculiar electronic structure at the same time endow them with interesting linear and nonlinear optical properties.^[5,6] Under irradiation with intense lasers, these nanomaterials demonstrate superior NLO response, such as OL and saturable absorption, which afterwards leads to successful demonstration of, e. g., ultrafast lasers based on saturable absorption of these CNT and graphene.^[7,8] Importantly, unlike semiconductor-based saturable absorbers (SAs) with limited operation bandwidth, these gapless carbon-based materials demonstrate ultra-broadband NLO response and ultrafast carrier cooling rate. These early studies have later stimulated growing effort in the investigation of the NLO properties of two-dimensional (2D) semiconductors immediately after their exfoliation by either chemical

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or physical methods in recent years. These 2D materials, such as MoS₂ and phosphorene, can be considered as mono- or few-layer stacked 2D crystal lattice of their bulk counterparts.^[9,10] Admittedly, the absence of subtle interlayer coupling in these 2D semiconductors, such as MoS₂ and phosphorene,^[11,12] contributes to the change in the electronic structure and eventually their distinct electrical and optical properties. Recent studies have confirmed that these 2D materials in the form of single or few-layered sheets indeed exhibit strong NLO response, manifesting strong saturable absorption or reverse saturable absorption. Similar to the carbon allotropes, the use of these 2D semiconductors as SA have also enabled the successful demonstration of Q-switching or mode-locking pulse generation at wavelengths from near infrared (NIR) to middle infrared (MIR) region. Unlike the 2D semiconductors, the broad band NLO response has been considered as an important but general character of gapless materials beyond graphenes, which has been confirmed in topological insulators (TIs) and semimetal materials in their LD form. The excellent NLO properties combined with the facile accessibility of these emerging LD materials makes them highly competitive for potential applications in pulse lasers and other photonic devices.

In the present review, we discuss the NLO properties and applications in pulse lasers of emerging LD materials with special focus on the non-classical plasmonic nanomaterials and 2D semiconductors. The nonlinear optical properties and their application as SA for pulse laser generation of fullerene as well as CNTs will not be discussed as it is already summarized in comprehensive reviews.^[13,14] This review begins with a general introduction of the size effect on semiconductor and plasmonic metal structures, as well as 2D materials including graphene and 2D semiconductors. We then focus on the NLO properties and ultrafast carrier dynamics of these materials assessed by different techniques. In the following sections we discuss the application of relevant LD materials as SAs in both fiber and solid state pulse lasers, as well as NLO devices. This review is finalized by a short comment on the current research status and possible future directions in the studying of the fundamental optical process and the applications of these materials.

2. Low-Dimensional Systems and their Linear Optical Properties

2.1. Size Effect in Semiconductor and Metal Nanostructures

It is well-known that absorption edge and photoluminescence of semiconductors shift to higher energy as a result of size reduction into a critical region. This size effect arises from the confinement of motion of electrons as the dimension is comparable or smaller than a critical size, i.e., Bohr radius of electrons (or holes, excitons).^[15,16] Semiconductors, in their 0D (zero-dimensional), 1D (one-dimensional) and 2D form, are known as quantum dots (QDs), quantum wires and quantum wells, in which the electronic density of states are distinct from their bulk counterparts, contributing to the change in optical bandgap and absorption characteristics. For instance, in the case of confinement in all three directions, i.e., QDs, the continuous band in bulk is reduced to atom-like energy levels.^[17]



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As the exciton Bohr radius is an intrinsic material properties related with dielectric constant (ϵ) and electron (or hole) effective mass (m_e , m_h), the quantum confinement effect in different

semiconductors can be simply divided into strong, intermediate and weak confinement regions.^[18,19] Semiconductors with large dielectric constant (and small m_e), such as lead-based QDs, usually possess a larger exciton Bohr radius (over 100 nm for PbTe) and therefore belong to the strong confinement case.^[20,21] In comparison, most oxide as well as halide semiconductors are characteristic weak confinement cases due to the relative large m_e and small ϵ .^[22] These parameters of semiconductors underpin their both linear and nonlinear optical properties and determine the fields of potential applications.

In metal nanostructures, the confinement of their conduction electrons in a dimension comparable with or smaller than the wavelength of light can result in the LSPR by the collective oscillation of electrons under light excitation. This LSPR effect endows nanoparticles of noble metals, such as gold and silver, with interesting optical properties attractive for diverse applications. The location of LSPR band depends on both intrinsic material properties such as free carrier concentration, and extrinsic properties, like size and shape.^[23,24] An increase in size can result in a spectral redshift. In addition to size, the optical spectral of the metal nanostructures with the same composition is also shape-dependent.^[25] For instance, nanorods usually exhibit two separated peaks located in high and low energy, corresponding respectively to the transverse and longitudinal resonance modes. The anisotropic optical properties of the nanorods is also prominent in the nonlinear region, as discussed in Section 3.3. Besides material properties, its LSPR band also varies with the dielectric function (i.e., refractive index n) of the surrounding environment, thus making it a versatile nanoscale platform for optical sensing.^[26–28]

On the other hand, semiconductors with heavy doping undergo cross-over to metallic state due to the presence of high concentration of conduction carriers (electron or holes). In recently years, there was growing interest in the exploiting of these non-classical plasmonic nanostructures based on, e.g., degenerate semiconductors.^[29] These semiconductor-based plasmonic materials include chalcogenide semiconductors and a large number of conducting oxides, as summarized in a recent review.^[30] Due to the much lower carrier density as compared to noble metals, these plasmonic materials usually demonstrate LSPR peak in the infrared spectral region, which, however, is not easily accessible by most noble metal plasmonic materials. As an alternative to metal based plasmonics, these materials have gained particular interest in NIR optics and photonics,^[30–32] as we will discuss in section 3.

2.2. 2D Materials and their Linear Optical Properties

2D materials are among the hottest research topics in the field of nanoscience since the discovery of graphene in 2004 by Geim et al.^[33] Graphene is built up from 2D hexagonal lattice of sp^2 hybridized carbons (Figure 1) which has been considered unstable in free space. In terms of electronic structure, graphene features two Dirac points and linear energy dispersion in momentum space. This unique electronic structure is characterized by the massless Dirac quasiparticles which leads to the extremely high carrier mobility promising for applications in electronics. Immediately after its discovery, study into

optical absorption found that its absorption for a monolayer is only related with fine structure constant, thus giving a constant absorption of 2.3% across a broad spectrum from visible to MIR region.^[34] Few-layer graphenes exhibit enhanced absorption that scales with the number of layers. The ultra-broad optical active bandwidth of graphene therefore facilitates broadband NLO response, as will be discussed in Section 3.

Following the study of graphene, van der Waals compounds, such as transition metal dichalcogenides (TMDs) and phosphorene (see Figure 1), have been successfully exfoliated and extensively investigated in recent years.^[35–37] These 2D semiconductors can be considered as natural quantum wells, thus featuring typical electronic structures with a 2D character. Of particular interest are the dichalcogenides of Mo and W, which characterizes bandgap in the range of 0.5–2.5 eV. Figure 1b shows the structure of MoS_2 in the 2H form (hexagonal, semiconducting); while MoS_2 and other TMDs can also stabilize with the 3R (rhombohedral, semiconducting) and 1T (tetragonal, metallic) structures. It has been confirmed from both the theory and experimental sides that these TMDs change from indirect to direct bandgap semiconductor in their monolayer form, resulting in the emergence of strong photoluminescence.^[38–40] Recent studies showed that electrons in the two distinct valleys can be selectively excited by circular polarized light.^[41,42] The valley polarized optical response has important implications for potential applications in spintronics and information technology. Another important 2D semiconductor is phosphorene, which is the monolayer version of black phosphorus (BP) (Figure 1), a high pressure allotrope of phosphorus with a layered structure.^[12,43] It has a direct bandgap of 2 eV (bulk ≈ 0.3 eV) for monolayer.^[44,45] Aside from TMDs and BP, a number of other 2D semiconductors have been reported recently.^[35–37] In the 2D material family, topological insulators of group V chalcogenide, such as Bi_2Se_3 , also adopt a 2D layered structure, with a large inter-layer van der Waals gap, as shown in Figure 1. TIs are such exotic matter which is characterized by a bulk bandgap with a Dirac-type surface metallic state similar to that of graphene.^[46,47] Up to the present, dozens of 3D TIs have been confirmed experimentally, including bismuth chalcogenide such as Bi_2Se_3 , and Bi_2Te_3 .^[48] Unlike previously reported 2D semiconductors such as MoS_2 , these gapless materials exhibit a broad linear absorption extended to the MIR region.^[49] Nevertheless, these 2D, van der Waals compounds have an important characteristic: they all can be easily accessed by mechanical exfoliation (ME) and liquid phase exfoliation (LPE).

3. Nonlinear Optical Properties in Emerging LD Materials

3.1. Nonlinear Optical Absorption and Experimental Methods

LD materials of both semiconductors and metals usually demonstrate stronger NLO response compared to their bulk counterparts due to the modulation of electronic structure as well as electronic density of states by structural low dimensionality. LD materials, as we discuss here, exhibit nonlinear absorption characterized by saturable absorption or reversible saturable

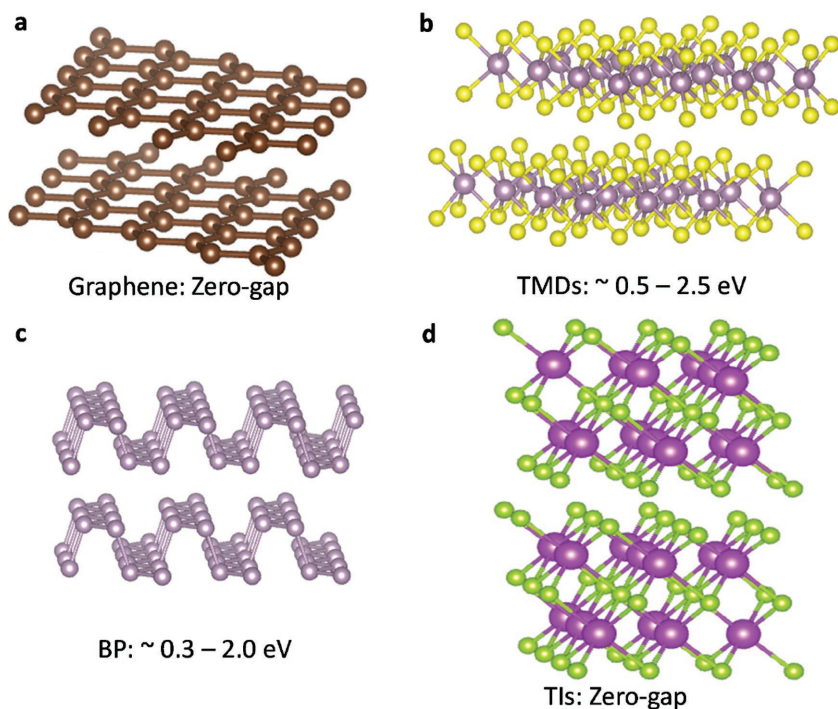


Figure 1. Structure of typical 2D materials characterized by a van der Waals gap. a) Graphene. b) Transition metal dichalcogenides (here as an example is the 2H structure of MoS_2). c) Phosphorene (black phosphorus). d) Topological insulator (here as an example is Bi_2Se_3). A two-layer stack is shown for each material so the van der Waals gap between the layers can be seen clearly. All these structures are generated by VESTA.

absorption, both of which have important implications for applications in different fields. Experimentally, NLO absorption can be examined by open (or close) aperture Z-scan^[50] and I-scan^[51] methods. From the published literature, it can be seen that the Z-scan method, which records transmittance of the laser as a function of the Z-axis distance of the sample (to the focus), is more often applied. In comparison, I-scan measurement is based on a compact all-fiber setup and it measures the samples' NLO absorption as a function of input power.

The NLO absorption has been described by different models. In the simplest expression, the absorption coefficient can be expressed as a function of laser intensity by:

$$\alpha(I) = \alpha_0 + \alpha_{\text{NL}}I \quad (1)$$

where α_0 and α_{NL} are the linear and nonlinear absorption coefficients, respectively. From a Z-scan or an I-scan measurement, both of the parameters can be experimentally determined. A more complex model has to be used for the examination of saturable absorption or reverse saturable absorption process by including the parameters of saturable (α_s) and nonsaturable (α_{NS}) absorption:

$$\alpha(I) = \alpha_s / (1 + I/I_s) + \alpha_{\text{NS}} + \beta I \quad (2)$$

where I_s is the saturation intensity and β is the reverse saturable absorption coefficient. This equation takes into consideration of both saturable absorption and reverse saturable absorption for a sample. If only saturable absorption is considered, the last

term of the above equation can be omitted for the calculation.

3.2. Plasmonic Semiconductor Nanocrystals

Degenerate semiconductors (such as Cu_{1-x}S and indium tin oxide (ITO)) have been considered as promising alternative plasmonic materials with their strong absorption band in the IR region, which can be far beyond the spectral region accessible by most noble-metal-based plasmonic materials.^[30–32] Due to their much lower carrier density as compared to metals, the plasmon frequencies of most conductive oxides such as ITO are located in the IR region, featuring a strong plasmonic absorption peak with a broad bandwidth. These NIR plasmonic materials have been given growing attention for electro-optical modulation,^[52] metamaterials,^[53] plasmon-based sensing,^[54] optical switches, etc.^[55–57] Furthermore, recent studies have showed that thin films of conductive oxides (such as ITO) demonstrate large optical nonlinearity in their epsilon-near-zero region (epsilon: dielectric permittivity).^[58,59] The change in the refractive index for ITO can reach 0.72 with a recovery time of 360 fs.^[58]

The nanoscale form of the above plasmonic semiconductor can be facily synthesized through a well-developed colloidal chemistry process, thus paving the way for their wide application in biomedicine and nanophotonics.^[60–63] The nanocrystals (NCs) of plasmonic semiconductors also exhibit large optical nonlinearity near their plasmonic resonance region. For instance, Guo et al.^[64] recently found that a thin film of colloidal Cu_{1-x}S NCs showed saturable absorption across a broad spectral region, which matches the linear plasmonic absorption. From Z-scan measurement, the saturable absorption component and the saturable intensity are 63.28% and 56.12 GW cm^{-2} , as shown in **Figure 2**. In addition, the colloidal Cu_{1-x}S thin film demonstrates a fast carrier relaxation rate, which can be decomposed into a fast sub-ps component originate from carrier–carrier scattering and a slow part due to carrier–phonon scattering (Figure 2f). The combination of these features is attractive for application as SAs in ultrafast lasers, as discussed in Section 4. Beside copper chalcogenide, degenerate oxide semiconductor NCs, such as ITO and AZO (ZnO:Al) exhibit an even broader saturable absorption region covering the entire NIR and part of the MIR band,^[65] making them a new class of NLO materials for a number of applications in nanophotonics.

It has been generally accepted that the saturable absorption in plasmonic NCs is associated with absorption by free carriers (electrons or holes).^[66,67] In conductive materials, the incident photons at any energy can generate electron–hole pairs, resulting in the thermalization of electrons and saturation of electronic state. A process known as Pauli blocking^[68] then becomes effective. This is different from the saturable

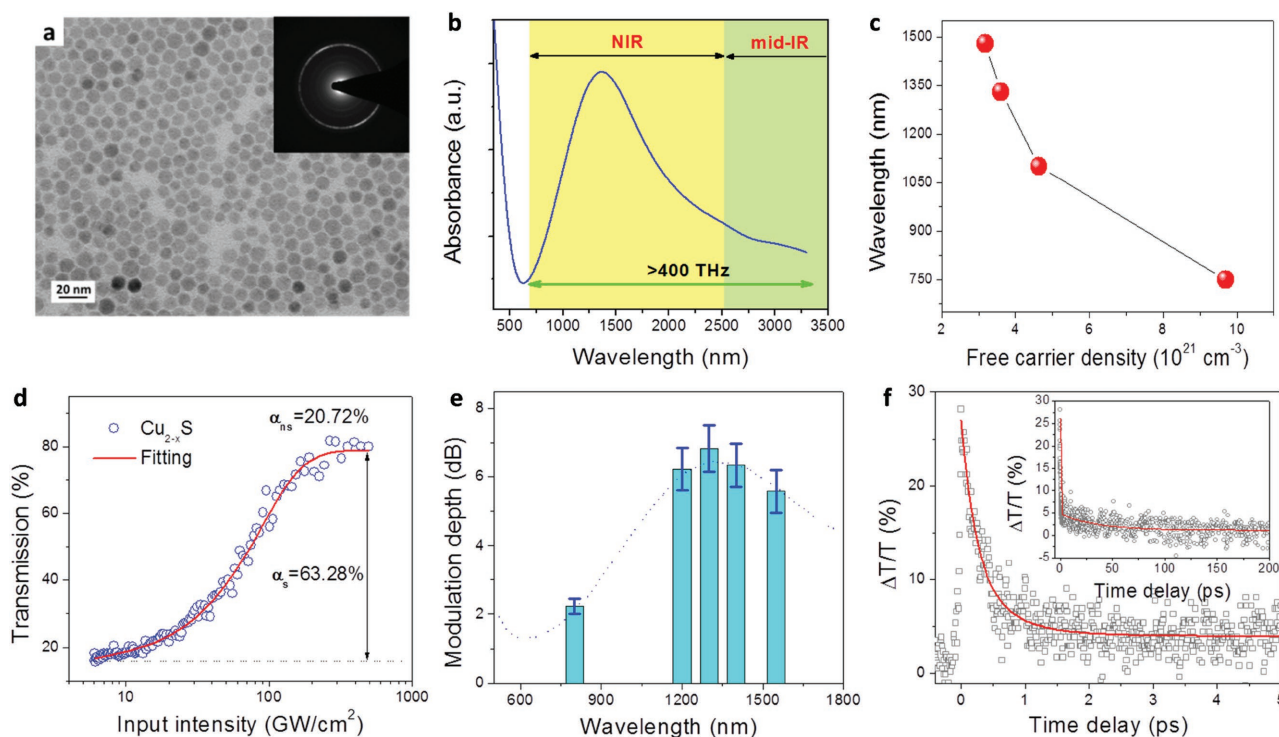


Figure 2. Linear and nonlinear optical properties of plasmonic Cu_{2-x}S NCs. a) Transmission electron microscope (TEM) image showing an average crystal size of 9 nm. The inset is the corresponding electron diffraction pattern. b) Linear absorption spectrum of the colloidal dispersion of Cu_{2-x}S NCs, showing a broad absorption band in the infrared region. c) The dependence of absorption peak wavelength on the carrier concentration calculated by the Drude model. d) NLO transmission curve (blue circle) recorded at 1300 nm by the open-aperture Z-scan technique. Fitting the experimental data with Equation (2) (shown as the red line) gives a modulation depth (α_s) of 63.28% and a non-saturable absorption (α_{NS}) of 20.72%. e) Wavelength dependence of modulation depth. A close match can be found between the linear absorption (shown as dotted line) and the modulation depth. f) Transient absorption curve of the Cu_{2-x}S plasmonic NCs at 1300 nm. Inset is the decay curves recorded up to 200 ps. The red curve is obtained by fitting using bi-exponential decay function. Reproduced with permission.^[64] Copyright 2016, American Chemical Society.

absorption in semiconductor materials, where electrons from the valence band are promoted to the conduction band by excitation with photons of sufficient energy and fluence, leaving the upper state filled such that the semiconductors are not able to absorb more photons due to the Pauli exclusion principle.^[69]

3.3. Assembled Plasmonic Nanostructures

The assembly of plasmonic anisotropic nanostructures (e.g., nanorods) can be used to generate polarization-dependent linear and NLO response. In general, stronger interaction is expected when the polarization direction of the incident light is parallel with the long axis of the nanorods. In oxide glasses containing aligned Ag nanorods (diameter: 20 nm, aspect ratio $\approx 2-4$) prepared by thermal drawing,^[70] reverse saturable absorption was observed with a clear optical anisotropy. From Z-scan measurement, the NLO absorption coefficient and the refractive index are found to be $\beta = 0.82 \text{ cm GW}^{-1}$ and $n_2 = -1.5 \times 10^{-4} \text{ cm}^2 \text{ GW}^{-1}$, respectively, for laser polarization along the long axis of the Ag nanorods. Both of these values are much higher than the results observed for laser polarization perpendicular to the Ag nanorods.

In contrast to oxide glass containing aligned Ag nanorods, a thin film of aligned Au rod prepared by Zhang et al.^[71]

demonstrates saturable absorption together with anisotropic NLO response. Their results clearly indicated the dependence of the nonlinear absorption coefficient and saturable intensity on the crossing angle (defined as the angle between the laser polarization and long axis of Au nanorods). As shown in Figure 3, the NLO response (β) reaches a maximum value when the crossing angle approaches 0° . This anisotropic NLO character is also manifested in the carrier relaxation measured by pump-probe experiment. The maximum transient transmittance change is observed for laser polarization parallel with the Au nanorods, when linear absorption reaches the maximum value.

3.4. Graphene and Related Materials

NLO absorption has been widely observed in sp^2 carbon-based materials such as carbon black solution (ink), C60 (and C70),^[5] CNTs^[14] and carbon dots in their liquid dispersion or blend with a polymer resin.^[72,73] Most of these systems demonstrate OL behavior; that is, reverse saturable absorption which can be used for laser protection, while CNTs of different form showed excellent saturable absorption which have been exploited further for use as SA for mode-locking pulse generation.^[7,74-76] In addition, the gapless character of these carbon nanomaterials

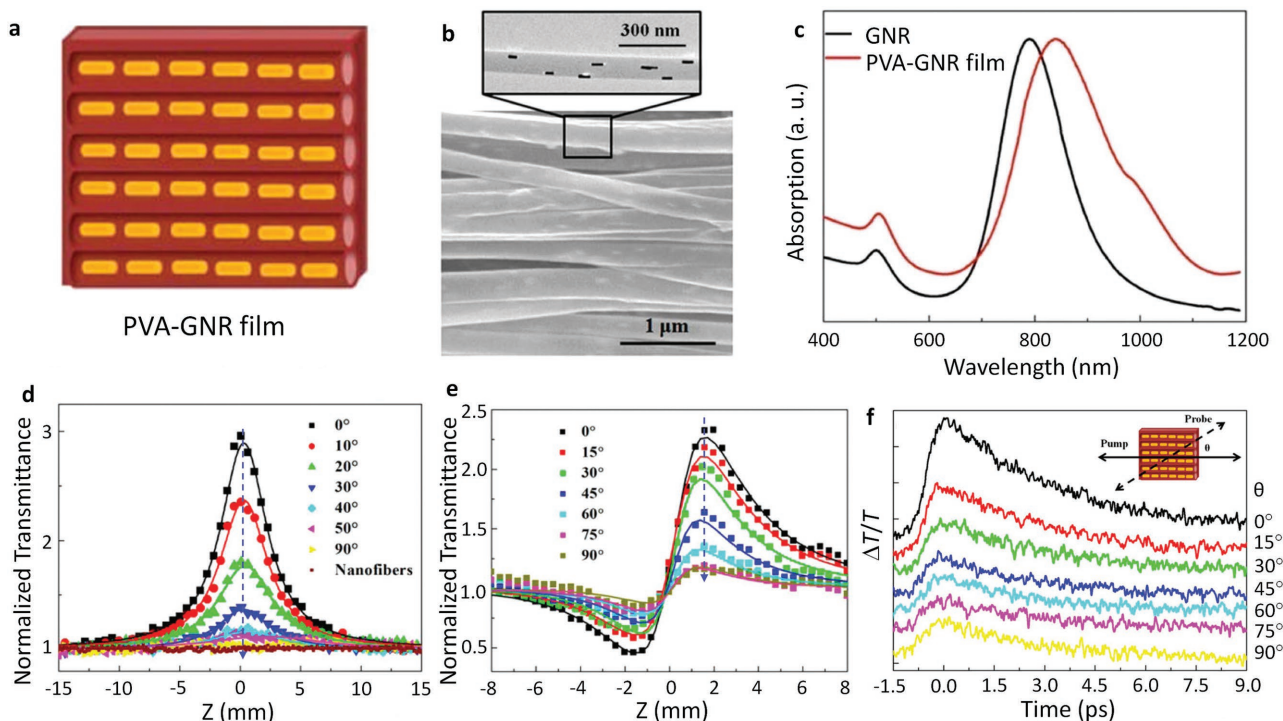


Figure 3. Polyvinyl alcohol (PVA)-gold nanorod (GNR) composite film containing unidirectionally aligned GNRs. a) Illustration of the microstructure of the PVA-GNR composite film. b) Scanning electron microscope image of the microstructure of the film prepared by electrospinning, showing clear unidirectional alignment of the GNRs. c) Absorption spectra of the GNRs and the PVA-GNR composite film. The resonance peak undergoes a small red shift for the composite film due to the change in the refractive index of the surrounding environment of GNRs in the composite as compared to bare GNRs. d) Open aperture and e) close aperture Z-scan measurement for the PVA-GNR composite film. Both groups of curves show a clear dependence of the normalized transmittance on the polarization angle of the pump laser (at 800 nm). f) Transient absorption curve recorded at 800 nm, where the polarization angle difference between the pump and probe beam (θ) varies from 0° to 90° . Reproduced with permission.^[71] Copyright 2015, American Chemical Society.

results in an ultra-broad saturable absorption band ranging from the visible to the IR region, which is inaccessible for commercial SA products.

Stimulated by these early observations, since the discovery of graphene investigation into NLO optical absorption followed naturally, aiming to exploiting its NLO properties for potential applications. The NLO absorption of graphene dispersion was first measured by Wang et al. using the Z-scan technique and a Q-switched Nd:YAG ($\text{Y}_3\text{Al}_5\text{O}_{12}$) laser at 1064 nm and 532 nm for excitation.^[77] OL behavior at both the visible (532 nm) and NIR (1064 nm) was observed as the pulse energy density exceeded 1 J cm^{-2} (pulse duration 6 ns), suggesting the broad band nature of the NLO response. Similar to other sp² carbon nanomaterials, the origin of OL of graphene dispersion was ascribed to laser-induced solvent bubbles as well as microplasmas. In other words, the solvent plays an important role in the laser-matter interaction and indeed solvents with different surface tension showed different OL threshold energy due to different bubble size. The authors therefore argued that NLO scattering (by the bubbles) was responsible for the observed OL, in addition to the widely accepted OL mechanisms such as multiphoton absorption.^[1,2] The strong solvent/matrix effect on the NLO absorption was also observed by Lim et al.^[78] for a dispersed single-layer sub-stoichiometric graphene oxide (sub-GOx) in both solvent and polymer matrix, as shown in Figure 4. However, the solvent effect was interpreted in terms

of the heavy atom effect,^[79] in which heavy atoms promote relaxation of the singlet excited state to triplet state due to spin-orbit coupling. Rather than nonlinear scattering, the OL was explained by the excited state absorption by the laser generated excitons of highly localized nature based on the results of nanoseconds pump-probe measurement (Figure 4). In addition to graphene, OL is observed in dozens of composites based on graphene and graphene oxide.^[80–82] Depending on the composition, the mechanisms of OL can be in general ascribed to nonlinear absorption, such as excited-state absorption, as well as nonlinear scattering.

Graphene has also been found to show saturable absorption by open-aperture Z-scan measurement. The origin of saturable absorption for graphene is explained by photo-bleaching due to Pauli-blocking. Bao et al.^[83] examined the nonlinear absorption of CVD (chemical vapor deposition)-grown graphene layers using a mode-locked fiber laser with pulse duration of 1 ps at 1550 nm. The results showed that α_s and α_{NS} are both dependent on the layer number, and the highest α_s of 66.5% with a saturation intensity of 0.71 MW cm^{-2} are found for graphenes of $\approx 2\text{--}4$ layers (Figure 5). More detailed examinations, however, found that the monolayer demonstrated a low modulation depth between 0.5% and 1.8% at wavelengths from 800 nm to 1500 nm,^[84–86] which is too low for stable mode-locking operation. By stacking few-layer graphenes the modulation depth can be enhanced, while the saturable intensity and

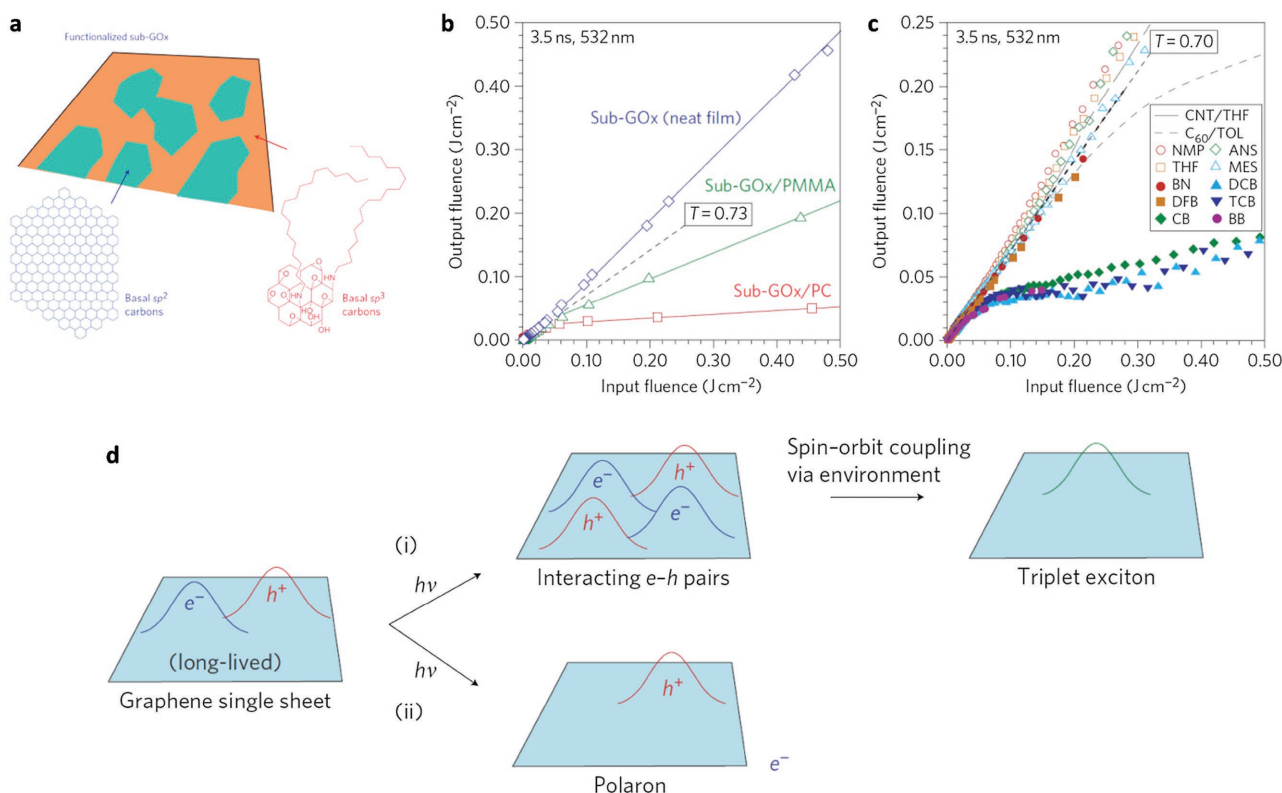


Figure 4. Optical limiting (OL) behavior of functionalized graphene dispersed in different solvents. a) Structure of the partially oxidized graphene sheet. The green domains are nanographenes which are separated by fully oxidized sp^3 carbon network. b, c) Effect of various solvents on the NLO absorption of the dispersed sub-GOx recorded at 532 nm (pulse duration 3.5 ns). b) Output as a function of input fluence for bare sub-GOx film, composite films of sub-GOx-PMMA and sub-GOx-PC. The transmittance for all films is 0.73. c) Output as a function of input fluence for sub-GOx dispersed in different solvents. The results for CNT in THF and C60 in TOL are shown for comparison. ANS, anisole; BB, bromobenzene; BN, benzonitrile; CB, chlorobenzene; DCB, 1,2-dichlorobenzene; DFB, 1,3-difluorobenzene; MES, mesitylene; NMP, N-methylpyrrolidinone; PC, polycarbonate; TCB, 1,2,4-trichlorobenzene; THF, tetrahydrofuran; TOL, toluene. T is 0.70. d) Mechanisms for optically induced absorption by sub-GOx. Following optical excitation, long-lived electron-hole pairs are generated. Further excitation can result in the formation of the localized states, i.e., (i) excitons and (ii) polarons (charged excited state) due to interaction. Reproduced with permission.^[78] Copyright 2011, Nature Publishing Group.

the nonsaturable loss are both compromised. Owing to the gapless character, saturable absorption across a broad spectral region has been observed for graphene layers deposited on solid substrate. For instance, Demetriou et al.^[87] confirmed saturable absorption from 1150 nm to 2400 nm with saturable intensity ranging from 1.9 to 4.5 GW cm^{-2} by using a fs (femtosecond) laser source. The irradiance independent nonlinear refraction is on the order of 10^{-7} cm^2 W^{-1} , a value 8 orders of magnitude higher than that of any dielectric material. However, the saturable fluence is much smaller at longer wavelengths (14 μJ cm^{-2} at 1500 nm compared to 66 μJ cm^{-2} at 800 nm), suggesting more favorable use in the MIR region.

For a deeper insight into the microscopic mechanisms of NLO absorption, ultrafast spectroscopy has been employed to study the carrier dynamics in time scales down to sub-100 fs with the help of Ti:sapphire fs lasers. The pump-probe experiment on graphenes synthesized by different methods has been performed by different groups and similar conclusions have been verified.^[88–93] In general, right after laser excitation a quasi-Fermi carrier distribution is established, which is followed by a slow relaxation processes including photoemission^[93] and scattering by the crystal lattice as well as defects.^[89]

Dawlaty et al. in 2008 measured the transient absorption of epitaxial graphene grown on SiC wafers.^[94] The measured curve can be fitted with a double-exponential decay function, giving a fast decay component of 70–120 fs range and a slow component of 0.4–1.7 ps, which corresponds to carrier–carrier scattering and carrier–phonon scattering, respectively. The rate of electron–phonon scattering is enhanced with an increase in crystal disorder, which can be confirmed by Raman spectroscopy. By the use of a terahertz probe pulse in the pump-probe experiment, this group later confirmed that the carrier recombination times are dependent on the carrier density, i.e., the pump density.^[92] In addition, a following investigation on graphene dispersion showed that this slow component is further delayed in the presence of solvent.^[95] The authors argue that solid substrate speed up the carrier scattering as compared with that of graphene liquid dispersion. Whereas substrate and solvent both can affect carrier relaxation, the inter-layer coupling may have played a subtle role in carrier cooling. Breusing et al.^[91] demonstrated that single-layer graphene exhibited a similar relaxation rate to that of a thick graphite layer^[90] as the interlayer coupling plays a minor role in carrier relaxation.

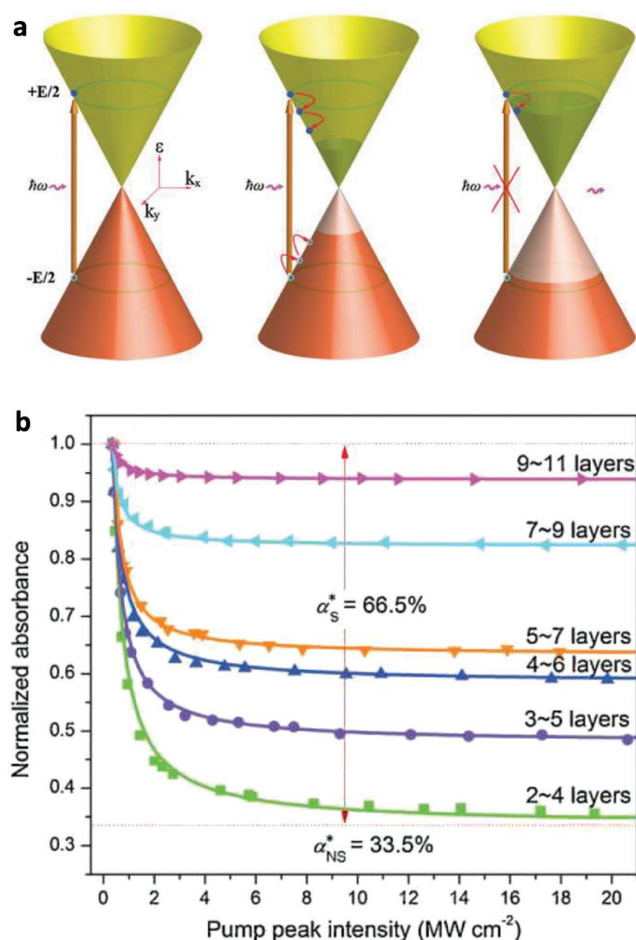


Figure 5. Saturable absorption by graphene. a) Schematic illustration of the mechanism of saturable absorption due to Pauli blocking. Left: Absorption of pumping light; middle: establishing of a Fermi-Dirac distribution of hot carriers; right: further absorption is blocked. b) Normalized absorbance as a function of pump power for graphenes of different thickness. Reproduced with permission.^[83]

3.5. 2D Semiconductors

3.5.1. Transition Metal Dichalcogenides

2D semiconductors have recently demonstrated intriguing optical properties, such as strong photoluminescence, as compared with their bulk counterparts. Their atomic thickness together with their peculiar electronic structure have also endowed them with enhanced NLO response. In terms of nonlinear absorption, these 2D semiconductors have demonstrated both saturable absorption and reverse saturable absorption depending on sample characteristics and measurement conditions. Wang et al.^[96] investigated the NLO absorption of dispersion of monolayer and few-layer MoS₂ by the Z-scan technique using a fs laser (800 nm) as pumping. As shown in **Figure 6**, for dispersion of MoS₂ of few-layer thickness prepared by sonication, saturable absorption was observed with a third order susceptibility $\text{Im}\chi^{(3)}$ in the order of 10^{-15} esu, and a smallest saturable intensity of 400 GW cm^{-2} . In addition, a comparison

with the same experimental conditions showed that a MoS₂ liquid dispersion was better than graphene in term of saturable absorption response (**Figure 6**).

The NLO absorption of MoS₂ is strongly dependent on the layer thickness. A comparative examination showed the dominating reverse saturable absorption upon sub-bandgap excitation for both dispersions of MoS₂ and MoSe₂ with a much smaller thickness.^[97] A more precise measurement reported by the same group confirmed the two-photon absorption of monolayer MoS₂ (synthesized by CVD) at sub-bandgap under fs laser excitation (1030 nm).^[98] Monolayer MoS₂ also demonstrates two-photon upconversion luminescence with a nonsaturation two-photon absorption coefficient of $(7.62 \pm 0.15) \times 10^3 \text{ cm GW}^{-1}$, which is three orders of magnitude higher than conventional semiconductors. Degenerate two-photon absorption is also observed for thin WS₂ (bandgap: $\approx 1.9 \text{ eV}$ for monolayer) films of different thickness (smaller than 41 layers) at 1030 nm (**Figure 6**)^[99] as well as WS₂ sheets dispersed in a PMMA matrix at 800 nm.^[100] Based on these experimental observations, the authors have tentatively ascribed the difference in NLO absorption between monolayer and few-layer MoS₂ to the variation of electronic structure. The saturable absorption of MoS₂ is energy dependent, and a transition from saturable absorption to reverse saturable absorption can occur at high laser energy due to the enhanced two-photon absorption and nonlinear scattering.^[101]

In addition to the liquid-phase-exfoliated and CVD-grown MoS₂ sheets, saturation absorption has also been observed by the Z-scan technique in MoS₂ synthesized by wet chemistry processes, such as the hydrothermal process.^[102] In addition to saturable absorption at above bandgap wavelength, however, these studies further confirmed saturable absorption at sub-bandgap region in the NIR, which was attributed to the presence of defects, edge states and surface dangling bonds that create energy levels within the bandgap. Particularly for 2D materials of atomic thickness, the absorption by these structural imperfections can be strong due to a large surface to volume ratio.^[103]

Ultrafast carrier dynamics following fs laser excitation have been studied to unravel more detailed photophysical process accounting for the NLO response in these 2D semiconductors.^[104–107] The different electronic structure in monolayer and few-layer MoS₂ results in distinct carrier relaxation dynamics as compared to their bulk counterparts when probed at the lowest exciton energy (A exciton).^[105] The transient absorption curves were fitted with a three-exponential decay function, giving three characteristic time component corresponding to the defect scattering ($\approx 3 \text{ ps}$), carrier-phonon scattering (80 ps) and direct recombination (800 ps). Due to the indirect-direct bandgap crossover in the monolayer, the recombination rate was enhanced by over 40 times for the monolayer. Furthermore, various forms of defects not only behave as scattering centers, but also facilitate nonradiative relaxation rather than radiative recombination. These results are in general agreement with a later observation in which the role played by defects was found to be the dominating mechanism with the involvement of the Auger process.^[108] There are, however, discrepancies in the carrier relaxation dynamics in the pump-probe experiments performed by different groups. A more

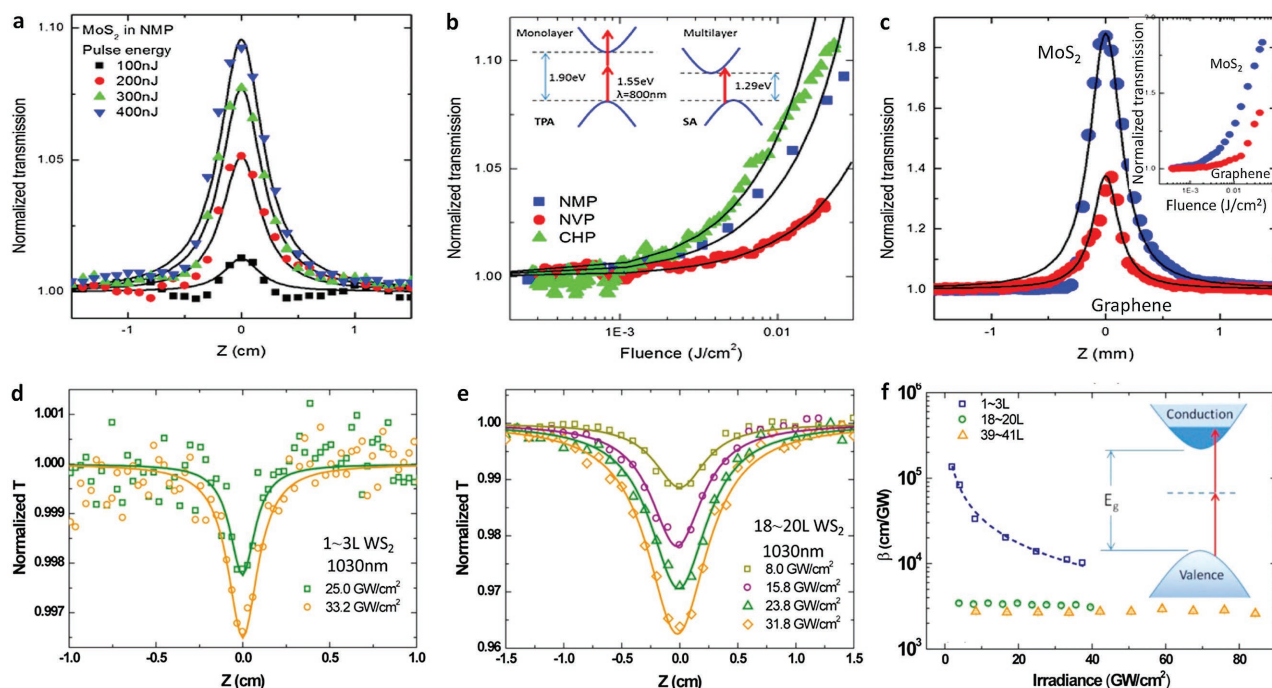


Figure 6. NLO properties of MoS₂ dispersions and thin films. a) Open aperture Z-scan measurement at 800 nm (100 fs) for MoS₂ dispersion in NMP. b) Normalized transmission as a function of laser fluence for MoS₂ dispersion in different solvents. The inset shows the optical transition for monolayer and multilayer MoS₂. c) Z-scan measurement for MoS₂ (T = 34.4%) and graphene (T = 16.5%) dispersions. The inset shows normalized transmission as a function of laser fluence. NMP, N-Methylpyrrolidone solvents; IPA, isopropyl alcohol; CHP, cyclohexylpyrrolidinone. Reproduced with permission.^[96] Copyright 2013, ACS. d, e) Open aperture Z-scan measurement at 1030 nm for WS₂ film (synthesized by vapor phase deposition) of varied thickness. f) Two-photon absorption (TPA) coefficient as a function of irradiance. Inset illustrates the TPA process. Reproduced with permission.^[99] Copyright 2015, American Chemical Society.

precise pump-probe experiment with better time resolution was performed for few-layer MoS₂ by Nie et al.^[107] at liquid nitrogen temperature by using a 10 fs pulse (centered at 669 nm with a bandwidth of 200 nm). Their results, however, indicated that carrier thermalization occurs within 20 fs and the subsequent carrier cooling by phonon scattering occurs on a time scale of 0.6 ps. This result is consistent with the most recent observations,^[104,109] suggesting that an intra-band relaxation time of sub-50 fs scale and inter-band transitions at ps time scales. This is similar to the two-step relaxation in graphene (see discussions above), which is regarded favorable for saturable absorption: the longer decay time ensures pulse self-starting and short relaxation allows for ultrashort pulse generation.

3.5.2. 2D Black Phosphorus: Phosphorene

Similar to 2D TMDs, phosphorene, the monolayer form of BP, also demonstrates strong nonlinear absorption in the form of either colloidal dispersion or thin film deposited on solid substrates. It has a variable direct bandgap from 0.3 eV to 2.0 eV depending on the number of layers. Therefore, phosphorene exhibits broad band absorption from the visible to the MIR spectral region, which is also manifested by the broadband NLO response. By Z-scan technique, it was first reported by Lu et al.^[110] that the dispersion of multi-layer black phosphorene

of 3–20 nm thickness dispersed in liquid exhibits strong saturable absorption at both 400 nm and 800 nm. Such dispersion therefore contained BP platelets with different bandgaps, featuring a broadened absorption spectrum. At 800 nm, the modulation depth and non-saturable absorption were found to be 12.4% and 1.9%, respectively. Changing of the dispersion solvent resulted in the variation of NLO coefficients, and a maximum modulation depth of 27.6% is observed for the IPA (isopropyl alcohol) dispersion. In a more recent report, Wang et al.^[111] examined the NLO absorption from 800 nm to 2100 nm. The results showed clearly a wavelength-dependent NLO response in the examined spectral region (Figure 7). The NLO absorption coefficient of BP dispersion is larger than that of graphene in the NIR region (1330–2100 nm; Figure 7d). Comparing BP sheets of larger lateral size, BP in the form of nanometer sized particles dispersed in liquid demonstrate the best modulation depth of 36% among BP-based SAs measured to date,^[112] and a low saturable intensity of only 3.3 GW cm⁻² at 800 nm. Besides BP dispersions, mechanically exfoliated BP layers deposited on a transparent substrate demonstrate even higher saturable absorption and lower saturable intensity.^[113] This strong saturable absorption combined with the broadband NLO response makes BP a promising SA material for pulse laser generation across the visible, NIR and MIR spectral regions.

Carrier dynamics in phosphorene as well as few-layer BP has also been intensively examined following its discovery in

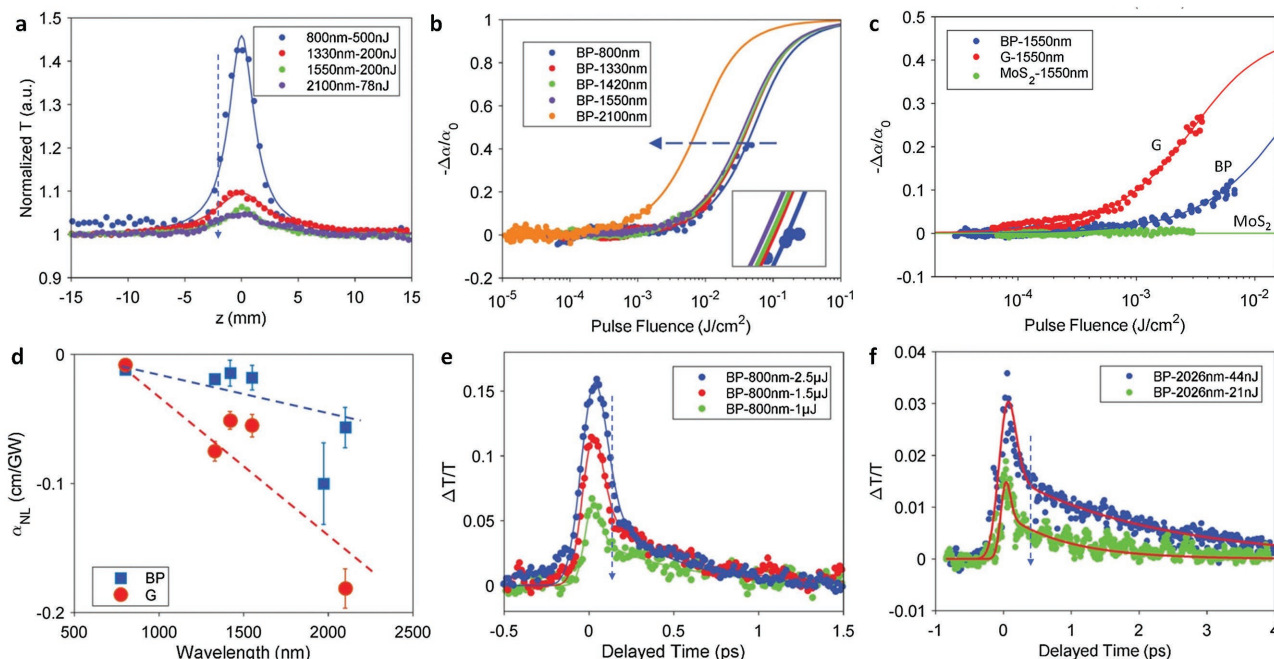


Figure 7. NLO absorption of BP dispersions. a) Open aperture Z-scan measurement for BP dispersion at different wavelengths from 800 to 2100 nm. b) Normalized differential absorbance as a function of laser fluence. c) Comparison of the normalized differential absorptivity for dispersions of BP, graphene and MoS₂ at 1550 nm. d) Calculated NLO coefficient for BP and graphene dispersions at different wavelength. Dashed lines serve as guide for eye. e, f) Transient differential transmission curves recorded for BP dispersion at e) 800 nm (e) and f) 2100 nm. Reproduced with permission.^[111] Copyright 2016, American Chemical Society.

2014. A pump-probe experiment was first performed on BP thin layers (around 80 layers) by Suess et al.^[114] by using a blueshifted 780 nm pulse as the pump and a 1560 nm pulse as the probe. The initial pump-induced photo-bleaching (in sub-ps timescales) is followed by the extended absorption, featuring two relaxation components of 180 ps and 1.3 ns. Similar to the carrier dynamics in other LD materials, the photo-bleaching can be ascribed to Pauli blocking. The enhanced absorption following the photo-bleaching can be ascribed to the Drude response of the pump-induced carriers, which showed strong polarization dependence related to the structural anisotropy of BP. This finding is in line with a recent angle-resolved transient reflectivity spectroscopy measurement.^[115] A pump-probe experiment was later performed on BP dispersions containing BP sheets with smaller thickness (<10 layers) and better time resolution (Figure 7e,f). In contrast to the temporal enhanced absorption discussed, in a timescale of less than 6 ps the BP dispersion exhibited a clear photo-bleaching, which decays rapidly with a fast and a slow time components of 360 fs and 1.36 ps, respectively.^[111] The relaxation rate is higher than that of graphene under the same measurement across the NIR and MIR regions.

3.6. Zero-Gap Materials: Topological Insulators and Dirac Semimetals

Compared with semiconductors, in which the infrared response is limited by the presence of a bandgap, gapless materials naturally exhibit broad-band NLO response, as has

been confirmed recently in some TIs^[116–120] and Dirac semimetals.^[121] The NLO absorption of a TI (Bi₂Te₃) was first examined by Zhao et al.^[116] by using the Z-scan technique with a ps laser at 1550 nm. Due to its layered structure Bi₂Te₃ precipitated in the hydrothermal process has a sheet-like morphology with a lateral size of 200–1000 nm and thickness of around 20 nm. Measurements on Bi₂Te₃ nanosheets deposited on a quartz slide yield a high modulation depth of 95% and a saturation intensity of 0.48 GW cm⁻². A following study on Bi₂Se₃ by the same group reported a modulation depth of 61.2% and a high third order nonlinear refractive index in the order of 10⁻¹⁴ m² W⁻¹.^[117] Analogously, Zhang et al.^[118] showed recently that the alloy of Bi₂Se₃ with Bi₂Te₃ (Bi₂Se₂Te₂) also exhibits high modulation depth (61.9%) and small nonsaturable loss (4.5%), which is appealing for application as SAs for mode-locking pulse generation.

Dirac semimetals (such as Cd₂As₃), which can be regarded as a 3D version of graphene according to electronic structure,^[122] have demonstrated similar ultra-broadband NLO absorption. Therefore, such materials are also gapless with wavelength-independent broadband absorption extended to the MIR range, similar to that of graphene. As demonstrated in a recent report of Zhu et al.,^[121] ultrafast NLO response with a characteristic fast time constant of 100–500 fs and slow component at the ps-scale was observed for a Cd₃As₂ film in the wavelength region from 1.6 to 4.7 μm, which has not been reached by other types of SAs. In addition, it was shown that doping can significantly speed up the carrier relaxation process due to the additional scattering by impurity centers.

4. Pulse Laser Generation Enabled by SAs based on LD Materials

4.1. SAs and Pulse Laser Technologies

Saturable absorbers have played an important role in the development of modern short-pulse lasers which are widely applied in fields from industrial processing to fundamental research. Compared with other types of modulators in pulse lasers, SAs allow both Q-switched and mode-locked pulse with controllable pulse parameters to be produced without resorting to the complex and costly electrical or Kerr modulators.^[123] In pulse lasers, SA serves as the passive optical switch that triggers laser pulse output by either Q-switching or mode-locking. The lasing parameters, such as pulse duration and repetition rate, are determined by the modulation depth and recovery time of the SAs, as well as the laser cavity design and spectral region of operation.

Pulse lasers generally operate in two distinct modes: Q-switching and mode-locking. For Q-switching lasers, the Q-factor is low when saturation of absorption by the SA is not reached, resulting in the accumulation of the intra-cavity energy. This leads to the absorption saturation by the SA and the surge of the Q-factor, resulting in the generation of a single pulse with much higher peak power. The SA and the gain medium then recover, awaiting energy accumulation for the generation of next pulses. Q-switched lasers normally operate with a low repetition rate (kHz) and moderate pulse duration (μs to ns) with high pulse energy (μJ to mJ). In comparison, mode-locked lasers operate based on a different mechanism. In the presence of SA, only the “spike” component of the noise can survive and it is amplified by the laser gain medium inside the cavity, eventually leading to the formation of the pulse train after iteration of this process. Because the cavity roundtrip time is the same as the period of modulation, the phase of each pulse is locked for the mode-locking operation. Fundamental mode-locking is achieved when a single pulse is circulating in the

cavity, while it is referred to as harmonic mode-locking for an integer number of pulses in the cavity. Generally, mode-locking lasers demonstrate high repetition rate (MHz to GHz), short pulse duration (ps to fs) and lower pulse energy compared with Q-switched lasers.

In the 1960s, coloured glass filters^[124] and organic “reversibly bleachable” dye^[125] were first used as SAs for Q-switching pulse generation right after the birth of the first laser.^[126] Currently, commercial SAs are based on semiconductor quantum wells, known as SESAMs (semiconductor saturable absorber mirrors).^[127] The limited operation bandwidth combined with the high fabrication cost of SESAMs has stimulated the continuous search for versatile SAs with facile accessibility and ultra-broad operation bandwidth. Since the 1990s, nanomaterials, such as CNTs, have shown to be effective SAs for both bulk and fiber lasers operating at different wavelengths.^[13,14,76] In addition, non-classical plasmonic NCs as well as emerging 2D materials also have enabled the generation of pulse lasers in different operation conditions at wavelengths extendable to the MIR region, as discussed below. One of the most important advantages of these LD materials as SA is the broad bandwidth for saturable absorption, while commercial SESAMs are based on resonant nonlinearity with limited saturable absorption bandwidth.

Unlike commercial SESAMs that operate in reflection, nanomaterials based SA have been incorporated into fiber or solid state lasers using different technologies.^[51] The simplest approach to fabricate SA is to deposit the colloidal dispersion of the nanomaterials (i.e., NCs or exfoliated 2D layers) onto a flat substrate, e. g., quartz glass plate or mirrors. SAs fabricated by this simple process can operate in transmission or reflection and they have been used for solid state lasers. Specifically for fiber-based SAs, different methods have been reported to integrating the SA into the fiber laser system. For instance, the SA can be fabricated by deposition of the nanomaterial dispersion onto the fiber connector facet, or onto a tapered fiber, as shown in **Figure 8**. The dispersion can be also coated onto a

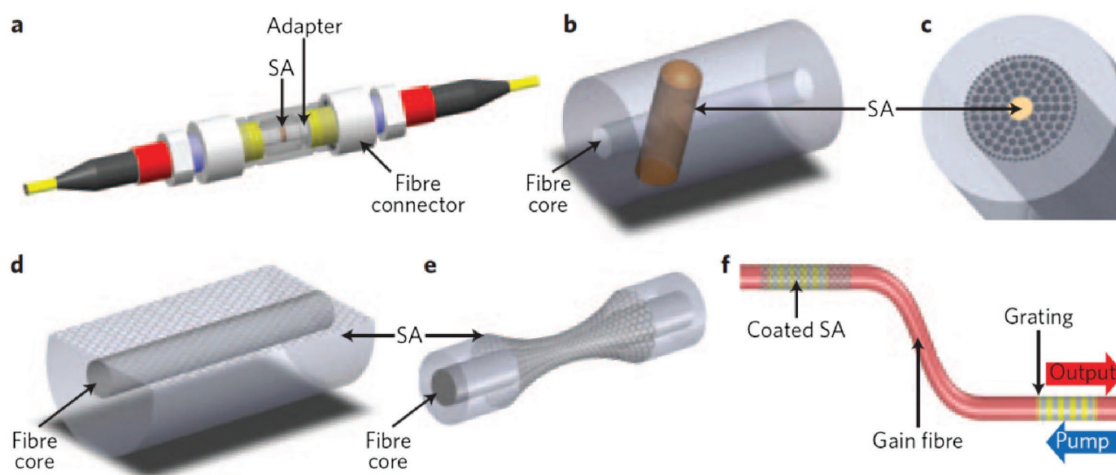


Figure 8. Different techniques for the integration of the SA into the fiber laser system. a) Sandwiched devices in which the SA material is deposited onto the fiber connector tip. b–f) SA deposited on to b) in-fiber microfluidic channels, c) photonic crystal fibers, d) a side-polished fiber core, e) tapered fiber and f) monolithic fiber laser. The SA shown here can be made from colloidal dispersion of LD materials or mechanically exfoliated thin sheets beside graphene. Reproduced with permission.^[7] Copyright 2013, Nature Publishing Group.

side-polished fiber based on the evanescent field interaction. The fibers employed as gain medium are often doped with different RE (rare earth) ions, such as Er^{3+} and Tm^{3+} , which give strong emission in the NIR and MIR.

4.2. Colloidal Plasmonic Semiconductor NC-Enabled Pulse Lasers

The conduction electrons in non-classical plasmonic crystals contribute to the strong and broad linear and nonlinear absorption in the IR and MIR spectral regions, as discussed previously. Recently, colloidal NCs of sub-stoichiometric Cu_{3-x}P with characteristic plasmonic absorption peaking at 1500 nm with a large bandwidth was found to exhibit strong saturable absorption. An SA fabricated based on the plasmonic colloidal NCs of Cu_{3-x}P (modulation depth 18.98%) has been employed for Q-switching pulse generation at 1500 nm based on a fiber laser cavity.^[128] The fiber laser consists of a 3 m Er-doped fiber and a 975 nm laser diode as pumping. The SA as Q-switcher was inserted between a polarization controller and the coupler. The laser output a shortest pulse of 5.094 μs and a highest pulse energy of 120.35 nJ at around 1562 nm with high stability.

The first mode-locking pulse laser generation based on plasmonic semiconductors was reported by Guo et al.^[64] using sub-stoichiometric Cu_{1-x}S colloidal NCs. Similar to Cu_{3-x}P , colloidal Cu_{1-x}S demonstrated a plasmonic absorption in the NIR region with a broad bandwidth of 400 THz, and its peak position can be finely tuned by chemical composition. Using the colloidal Cu_{1-x}S NCs, mode-locking pulse generation was realized by different fiber lasers at wavelengths from NIR to MIR.^[64] For the 1550 nm mode-locking pulse generation, the SA was fabricated by coating the Cu_{1-x}S NC dispersion onto the fiber (Er-doped fiber) based on the evanescent field interaction (Figure 9). A self-starting mode-locking pulse was generated at around 1562 nm with a high signal-to-noise ratio (50 dB), as shown in Figure 9. The fiber laser delivered a stable pulse train with a repetition rate of 7.28 MHz and a temporal pulse duration of 295 fs. A different setup was employed for demonstration of the pulse laser at 2.8 μm based on an Er-doped ZBLAN fiber. The SA in this laser was prepared by spraying the Cu_{1-x}S NC dispersion onto a reflective mirror. This laser generated stable Q-switching pulses centered at 2769 nm with a repetition rate of 66.4 kHz and average power of 126 mW. Beside fiber lasers, a solid state laser based on Yb:LuYAG crystal was also demonstrated by using a similar SA mirror operating in reflection. This laser output 1030 nm mode-locking pulse with a pulse duration of 7.8 ps. A more recent report from the same group showed that the Cu–Sn–S colloidal crystals can be also fabricated into an efficient SA with comparable performance.^[129] The 1550 nm fiber laser in which SA is integrated into the fiber by evanescent field interaction demonstrated stable pulse train output with a repetition rate of 4.99 MHz and a pulse duration of 923 fs.

Beside phosphide and chalcogenide plasmonic NCs, oxide plasmonic NCs synthesized by a colloidal chemistry route have also been fabricated into SA and enabled mode-locking pulse generation in the NIR region (1550 nm). These oxide-based SA

are promising for commercial applications due to its intrinsic high stability, high laser damage threshold, as well as facile processibility.^[65]

4.3. Graphene-Based-SA-Enabled Pulse Lasers

Due to strong saturable absorption, ultrafast response time (in the order of 100 fs) and ultra-broad NLO response bandwidth, graphene SAs have enabled pulse laser generation from visible to MIR range.^[7] The easiest process to fabricate graphene-based SA is to deposit a graphene layer onto the fiber connector, as shown in Figure 8. The first graphene SA based mode-locking laser was reported by Bao et al.^[83] based on an Er-doped fiber. The laser operated at 1565 nm with a pulse duration of 756 fs. Following this experiment, Sun et al.^[130] reported a fs laser using a graphene-based SA operating in a similar wavelength region. The SA film was made from graphene–PVA composite and it was inserted into the fiber connector. The mode-locking pulse demonstrated a pulse duration of ≈ 460 fs with a spectral width of 5.24 nm. The pulse duration was further reduced to the sub-200 fs region by the same group based on a similar laser system and the same graphene–PVA SA (see Table 1 for more technical details).

Using graphene-based SAs, a large number of studies have been devoted to develop pulse fiber lasers operating in the MIR range based on, e.g., Tm-doped fibers (see Table 1). Lasers in this wavelength region are of particular interest for applications in remote sensing, medicine and spectroscopy. A MIR mode-locking fiber laser using a graphene SA was first reported by Zhang et al. in 2012.^[131] The laser emitted a 1.94 μm pulse with a pulse duration of 3.6 ps using SA made by PVA film containing exfoliated graphenes. Based on similar Tm-doped fibers, the pulse laser employing the graphene SA have been reported by other groups later with shorter pulse durations down to 654 fs. Zhu et al.^[132] applied a different technique in the demonstration of a 2.8 μm laser using Er-doped ZBLAN fiber. Graphene was coated onto a mirror and the SA operated in reflection. The laser generated 42 ps pulse at 25.4 MHz with a pulse energy of 0.7 nJ. Table 1 summarizes the operation parameters of mode-locking lasers using graphene-based SA.

On the other hand, high repetition rate lasers based on a short cavity are highly desirable for optical communication and other relevant applications.^[133] Commercial SESAMs have to be used in free space which increases the system complexity. In comparison, graphene dispersed in solvent or its blend with a polymer resin (e.g., PVA) can be directly deposited onto fiber facet and serves as SA in fiber lasers (Figure 10). By using the Fabry–Pérot cavity and a short fiber cavity of only 10 mm, Martinez et al. reported a 10 GHz fundamental mode fiber laser operating at a central wavelength of 1562 nm based on an Er–Yb-coped phosphate glass fiber.^[134]

Following the development of fiber lasers using graphene based SA, solid state lasers also have been successfully demonstrated operating in the IR region. In contrast to fiber lasers, the SAs are made by deposition of graphene onto a mirror, such that the SA operates by reflection in most reported solid state lasers. The first solid state laser using graphene SA was based on a Nd:YAG ceramic,^[135] using graphene coated mirror

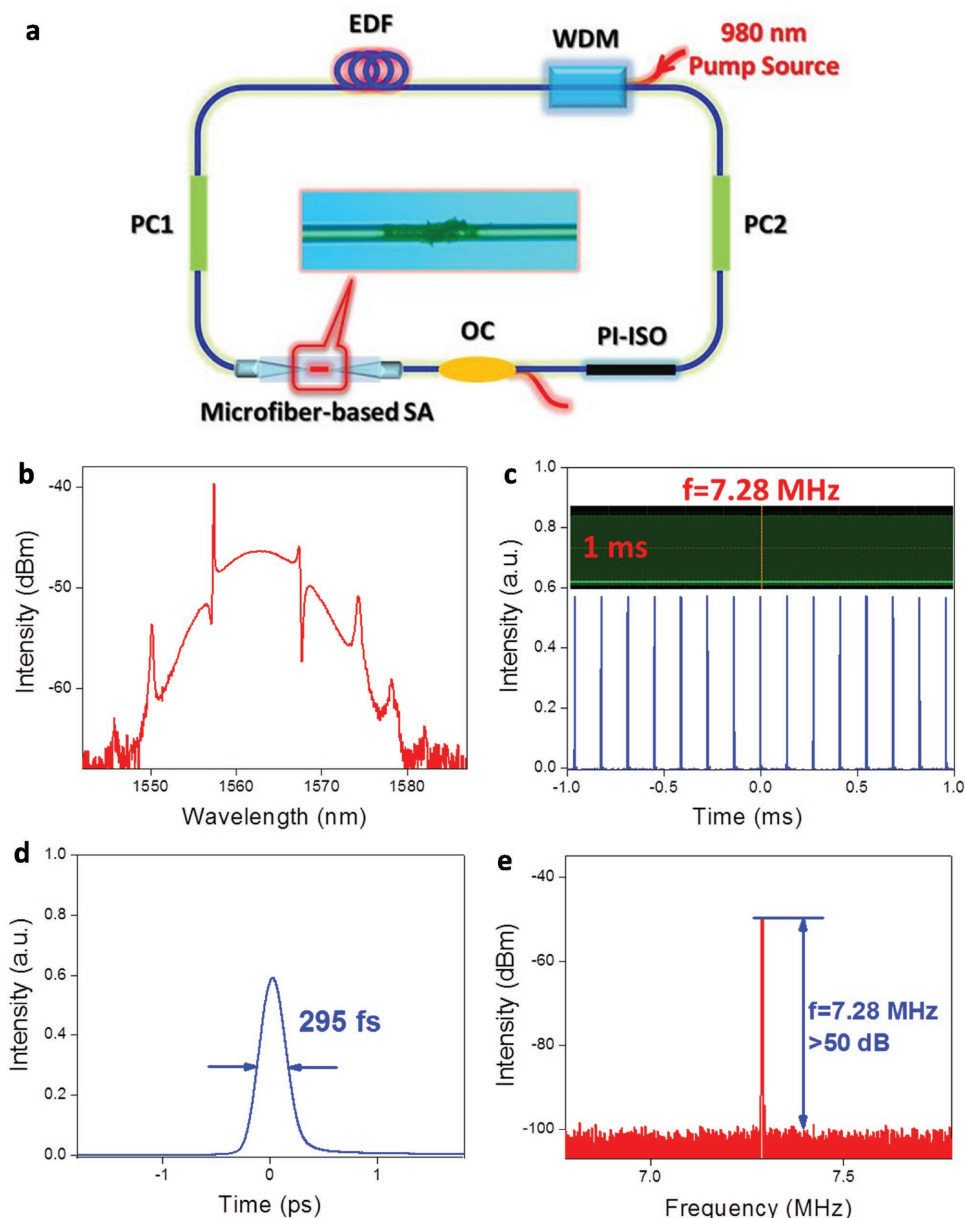


Figure 9. Mode-locking pulse generation at 1550 nm using plasmonic Cu_{2-x}S NCs as SA based on an Er-doped fiber. a) Schematic illustration of the microfiber-based plasmonic NCs SA mode-locked fiber laser. EDF, Er-doped fiber; PC, polarization controller; PI-ISO, polarization-independent isolator; OC, optical coupler (10%). b) Typical mode-locking optical spectrum. c) mode-locking pulse train. d) autocorrelation trace. e) the RF (radio-frequency) optical spectrum at the fundamental frequency. Reproduced with permission.^[64] Copyright 2016, American Chemical Society.

as SA. The laser pulse duration was 4 ps with an average output power of 100 mW. Later, graphene SAs have allowed pulse laser generation in the IR region using RE-doped laser crystals as gain medium, including Nd:YVO₄,^[136] Yb:YGG (Y₃Ga₅O₁₂), Yb:LuGG (Lu₃Ga₅O₁₂), Yb:CNGG (Ca₃Nb_{1.5}Ga_{3.5}O₁₂)^[137] and Nd:LuVO₄.^[138] The shortest laser pulse generated based on the graphene SA reached by solid state lasers was based on a Yb:CaYAlO₄ crystal,^[139] which generated a 30 fs pulse with a pulse width of 50 nm.

In addition to the lasers using RE-doped fibers and crystals, IR and MIR mode-locking lasers based on transition metal doped crystals have also been developed. For instance,

A Ti: sapphire mode-locking pulse laser was demonstrated by Baek et al,^[84] which generate ultrashort pulse of 63 fs (at 800 nm) with maximum power of 480 mW. A 68 fs laser was reported by Canbaz based on Cr:LiSAF(LiSrAlF₆) operating at 850 nm.^[140] On the other hand, graphene SAs have also enabled several MIR pulse lasers based on Cr²⁺ doped chalcogenide crystals such as ZnS.^[141–143] These lasers operated in the 2.5 μm region and the shortest pulse reached 41 fs based on a Cr:ZnS crystal.^[143]

Graphene SAs have also been used to replace the commercial SESAMs in ultrafast vertical-external-cavity surface-emitting lasers (VECSELs), sometimes referred to as semiconductor

Table 1. Graphene-based SA enabled pulse lasers.

Type of Materials	Material characteristics		SA integration method	Laser properties ^{a)}					Years reported [ref.]	
	Fabrication method	Thickness (nm); number of layers (n)		Laser type	Gain medium	λ (nm)	T	f_{rep}		P_{max} (mW)
Gr.	CVD	2–4 layers	Fiber facet	Mode-locking	Er-doped fiber	1565	756 fs	1.79 MHz	2	2009 ^[83]
	LPE	–	Gr/PVA film	Mode-locking	Er-doped fiber	1559	460	19.9 MHz	–	2010 ^[130]
	Reduction of GO	–	Microfiber	Mode-locking	Er-doped fiber	1561.6	1.3 ps	6.99 MHz	–	2010 ^[147]
	Reduction of GO	–	SA mirror	Mode-locking	Nd:YAG crystal	1064	4 ps	88 MHz	100	2010 ^[135]
	LPE	–	Gr/PVA	Mode-locking	Er-doped fiber	1525–1559	≈1 ps	8 MHz	≈1	2010 ^[148]
	Reduction of GO	≈1 nm; 1 layer	Fiber facet	Q-switching	Er-doped fiber	1566.35	3.7 μs	65.9 kHz	1.1	2010 ^[149]
	CVD	1 layer	Fiber facet	Mode-locking	Er-doped fiber	1561	1.23 ps	2.54 MHz	3.0	2011 ^[150]
	CVD	1 layer	Quartz slide	Mode-locking	Cr:Mg ₂ SiO ₄ crystal	1220–1270	94 fs	74.65 MHz	230	2011 ^[151]
	CVD	1 layer	Quartz slide	Mode-locking	Yb:KLuW crystal	1047	160 fs	92.52 MHz	160	2012 ^[86]
	CVD	1 layer	SA mirror	Mode-locking	Ti:Al ₂ O ₃	800	63	99.4 MHz	480	2012 ^[84]
Commercial graphene dispersion	LPE	Few-layer	SA mirror	Mode-locking	Er,Yb-codoped fiber	1576	865 fs	9.67 GHz	–	2012 ^[134]
	Commercial graphene dispersion	–	SA mirror	Q-switching	Er-doped ZBLAN fiber	2783	1 μs	37 kHz	62	2013 ^[152]
	LPE	–	Gr/PVA	Mode-locking	Tm-doped fiber	1940	3.6 ps	6.46 MHz	≈2	2013 ^[131]
	CVD	1 layer	SA mirror	Mode-locking	Cr:YAG crystal	1516	91 fs	85 MHz	107	2013 ^[85]
	CVD	1 layer	Gr. on CaF ₂ substrate	Mode-locking	Cr:ZnSe crystal	2500	226 fs	77 MHz	80	2013 ^[141]
	CVD	Few-layer	SA mirror	Mode-locking	Cr:ZnS	≈2400	189 fs	46 MHz	890	2014 ^[142]
	CVD	Few-layer	SA mirror	Mode-locking	Cr:ZnS	≈2400	41 fs	108 MHz	250	2014 ^[143]
	CVD	>5 layers	Fiber facet	Mode-locking	Yb-doped fiber	1035	6.5 ns	16.29 MHz	–	2014 ^[153]
	–	–	–	–	Er-doped fiber	1564	870 fs	19.3 MHz	–	–
	–	–	–	–	Tm:Ho-doped fiber	1908	65 ns	1.82 MHz	–	–
GO	CVD	Few-layer	SA mirror	Mode-locking	AlAs/GaAs Bragg mirror	1030	353 fs	26.5 GHz	10 W	2014 ^[144]
	CVD	1 layer	SA mirror	Mode-locking	Yb:CaYAlO ₄ crystal	1068	30 fs	113.5 MHz	–	2016 ^[139]
	LPE	1–3 layers	Gr./PMMA film	Mode-locking	Er-doped fiber	1550	326 fs	4 MHz	–	2016 ^[154]
	CVD	4–6 layers	SA mirror	Mode-locking	Er-doped ZBLAN fiber	2784.5	≈42 ps	25.4 MHz	18	2016 ^[132]
	Hummers method	≈1 nm; 1 layer	GO/PVA	Q-switching	Pr ³⁺ -doped fluorozirconate fiber	637.5	554 ns	195.3 kHz	3.5	2016 ^[155]
r-GO	LPE	–	GO/PVA film	Q-switching	Tm-doped double-clad fiber laser	1950.27	1.5 μs	83.2 kHz	730	2016 ^[156]
	–	–	rGO on PCF	Mode-locking	Er/Yb co-doped double-clad fiber	1564.9	791 fs	11.73 MHz	1140	2016 ^[157]

^{a)} λ (nm) is the central operation wavelength, T is pulse duration, f_{rep} is the (highest) repetition rate and P_{max} is the maximum operation power of the laser. Gr., graphene; GO, graphene oxide; r-GO, reduced graphene oxide; PCF, photonic crystal fiber.

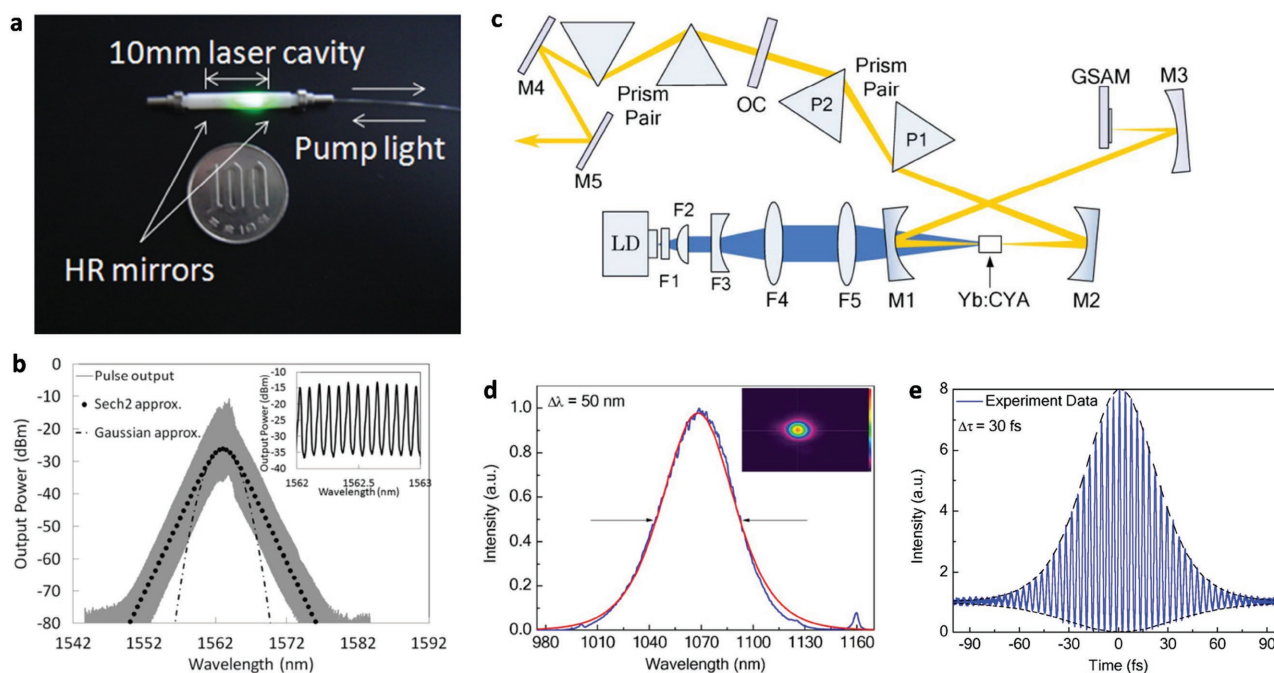


Figure 10. Pulse lasers using graphene based SAs. a, b) 1550 nm fiber laser operates at 10 GHz. a) Photograph of the fiber laser based on Fabry-Pérot using a Er-Yb codoped gain fiber. The green emission is due to the upconversion by Er ions. b) Optical spectrum of the fiber laser. The dotted curve is the fitting by sech² function. Inset shows the optical spectrum at the peak wavelength region. The spacing between each mode is 0.08 nm, corresponding to a repetition rate of 9.67 GHz. Reproduced with permission.^[134] Copyright 2012, AIP. c–e) Solid state mode-locking laser for 30 fs pulse generation based on graphene SA. c) Laser setup using graphene saturable absorber mirror (GSAM). F1 – F5 are different lenses; M1, M2 M3 are concave mirrors; M4 and M4 are folding mirrors. The laser gain medium is a Yb:CaAlO₄ crystal. d) Spectrum of the mode-locked pulse. The inset is the beam profile recorded after the output coupler. e) Autocorrelation trace of the mode-locked pulse at 1070 nm. Dotted curve is the fitting using the sech² pulse shape function. Reproduced with permission.^[139] Copyright 2013, Optical Society of America.

disc lasers (SDLs). SDLs have broad applications in fields such as optical communication, optical microscopy and super-continuum generation.^[144–146] Zaugg et al.^[145] employed a graphene-integrated distributed Bragg reflector as SA, where a spacer is sandwiched between a reflection mirror and the graphene. By changing the thickness of the spacer the electric field intensity enhancement can be adjusted between 0 and 400%. This graphene SA was used to mode-lock different semiconductor gain chip emitting from 935 nm to 981 nm, with pulse duration down to 466 fs and repetition rate up to 2.48 GHz. Later, a similar SDL operating at 1030 nm was demonstrated by Husaini et al.^[144] using graphene SA, giving a record output power of 10 W and pulse duration of 353 fs.

4.4. 2D Semiconductor-Enabled Pulse Lasers

4.4.1. MoS₂-Based SA Enabled Pulse Lasers

Following the successful demonstration of pulse laser using graphene-based SA, there has been growing number of demonstrations on the pulse laser using SAs based on 2D semiconductors of dichalcogenides of W or Mo. The presence of edge state as well as the defect state contributes to the SA in the NIR and MIR region, allowing the demonstration of fiber lasers at wavelengths far longer than that of the fundamental exciton absorption wavelength (for monolayer MoS₂, 1.8 eV, i.e.,

689 nm). Generally, the SAs based on these 2D materials can be fabricated using similar techniques applied for graphene-based SAs. Either the liquid dispersions or its blends with polymer (i.e., PVA) can be deposited or coated onto fiber facet or surfaces to allow stable pulse output. Wang et al.^[102] first demonstrated solid state pulse laser using the MoS₂-based SA employing bulk Q-switched cavity. Pulse lasers at wavelengths of 1.06 μm, 1.42 μm and 2.1 μm were generated based on laser crystals of Nd:GdVO₄, Nd:YGG and Tm:Ho:YGG, respectively. As presented in Figure 11, the maximum output power of ≈230 mW is reached by the 1.06 μm (based on Nd:GdVO₄) laser under pumping power of 2.7 W. Due to the presence of defects and edge states, saturable absorption still occurs at wavelengths far below the band gap of MoS₂, thus enabling the extension of pulse output to the MIR region. The first fiber laser using MoS₂-based SA was reported at the same year by Woodward et al.^[158] operating at 1068 nm. The operation wavelength for the MoS₂-based Q-switching fiber lasers was quickly extended by other groups to longer wavelengths up to 2 μm region.^[159] From the published results, solid state lasers based on RE-doped crystals deliver a higher average power and lower repetition rate compared with fiber lasers using similar MoS₂-based SAs.

After the successful demonstration of a Q-switching laser using MoS₂-based SA, mode-locking pulse generation was first realized by Zhang et al.^[101] based on a Yb-doped fiber laser operating at 1054 nm (peak wavelength) with a bandwidth of

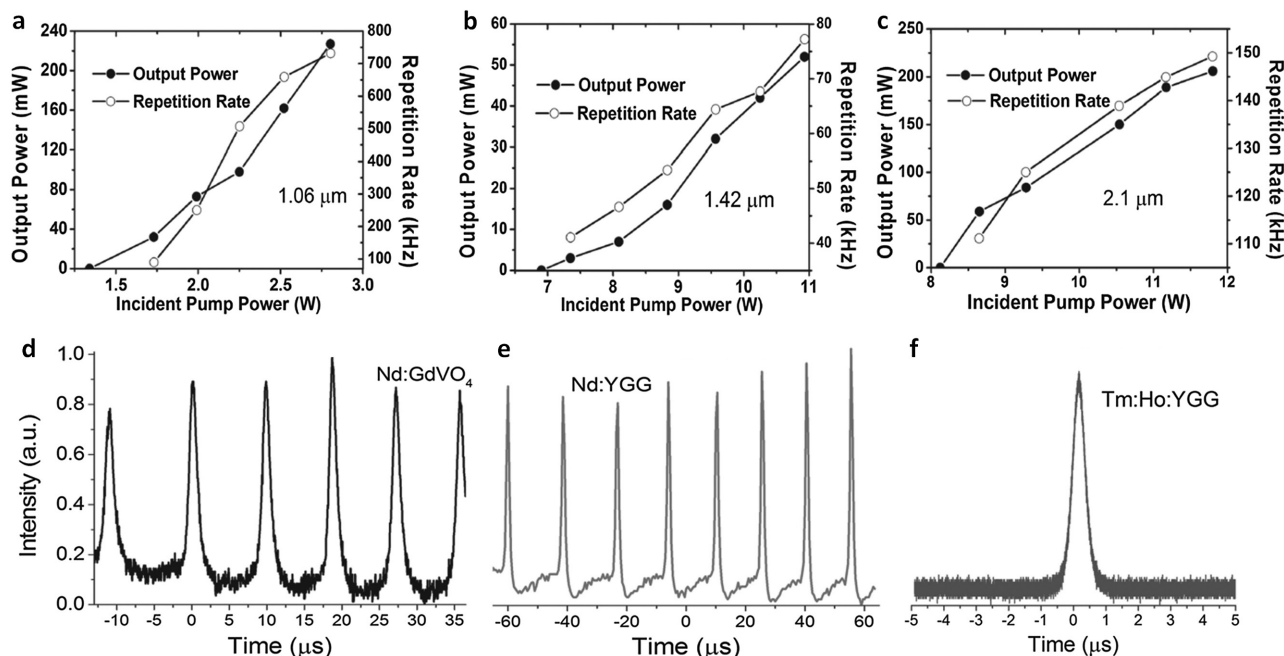


Figure 11. Q-switched solid state laser based on MoS₂ SA. a–c) Dependence of output power on the pump power for the solid state lasers operating at: a) 1.060 μm, b) 1.42 μm and c) 2.1 μm using gain medium of Nd:GdVO₄, Nd:YGG and Tm, Ho:YGG. d–f) Q-switched pulse train for d) Nd:GdVO₄, e) Nd:YGG and f) Tm, Ho:YGG lasers, generating pulse durations of 970 ns, 729 ns and 410 ns, respectively. Reproduced with permission.^[102]

2.7 nm. The SA employed in this laser was made by deposition of the liquid phase exfoliated MoS₂ onto the fiber connector. Extending the operating wavelength using the same MoS₂-based SA was reported later by the same group based on an Er-doped fiber. This mode-locking laser produced stable 710 fs pulse at a central wavelength of 1569.5 nm with a repetition rate of 12.09 kHz.^[160] Wavelength-tunable laser operation was also successfully demonstrated by using a MoS₂-based SA, which generated continuously tunable pulses from 1535 nm to 1565 nm.^[161] Up to the present, mode-locking fiber lasers employing MoS₂-based SA all adopted a unidirectional ring cavity design. In contrast, mode-locking pulse generation still has yet to be realized in solid state lasers based on a bulk-cavity, possibly due to the high nonsaturable loss of such SAs, which favors a Q-switching operation. From the published results, the shortest pulse duration reached is 710 fs^[160] and the highest repetition rate is 2.5 GHz.^[162] Further improvements of these parameters are still expected by optimization of the SA and laser cavity design in the future.

4.4.2. Other 2D-TMD-Based Pulse Lasers

2D semiconductors of other TM dichalcogenides, such as MoSe₂, WSe₂ and WTe₂, exhibit similar NLO absorption and SAs based on these materials have also been employed for both Q-switching and mode-locking pulse lasers. In terms of NLO properties, the major advantage of these heavier 2D TMDs is the smaller bandgap, which favors pulse laser generation at much longer wavelengths, especially in the MIR region. Both Q-switching and mode-locking pulse generation have been successfully demonstrated in the past two years using

TMDs other than MoS₂. For instance, using a MoSe₂-based SA, a Q-switching laser was demonstrated at wavelengths of 1060 nm, 1566 nm and 1924 nm using Yb, Er and Tm doped fibers, respectively.^[163] For these lasers, the SA was a MoSe₂-PVA film which was integrated between two fiber patch cords. Using a similar MoSe₂-PVA film as the SA,^[164] Koo et al. in a recent report showed the generation of a fs pulse with a high repetition rate of 3.27 GHz based on an Er-doped gain fiber. Other tungsten-based TMDs have also been fabricated into similar SAs and have enabled pulse lasers at different wavelength, as listed in Table 2.

Along with the rapid advances in IR lasers based on TMD SAs, there is also continued interest in the development of fiber lasers operating at visible wavelength based on 2D TMD-based SAs, by making use of resonant optical nonlinearity above the fundamental exciton wavelength region (1.8 eV for monolayer MoS₂, i.e., 689 nm). At this wavelength graphene has been considered not suitable for application as SA because the saturation intensity was much lower at visible region.^[150] The relatively large bandgap and higher modulation depth in the visible range of mono- or few-layer TMDs therefore make themselves favorable SAs for pulse laser operating at shorter wavelengths. Luo et al.^[165] demonstrated a red-light pulse laser operating at 635 nm based on a Pr-doped fluoride fiber. As shown in Figure 12, 2D semiconductors of MoS₂, MoSe₂ and WS₂ can be all fabricated as TMD-PVA SA and integrated in a fiber connector. These SAs made from different 2D materials all enabled Q-switched visible laser pulse generation with minimum pulse duration from 207 to 240 ns. For the WS₂-based SA, the repetition rate increases from 240 to 520 kHz and the pulse duration changes from 200 to 650 ns when the pumping power increases from 140 to 210 mW.

Table 2. 2D semiconductor (including TMDs and BP) based SA enabled pulse lasers.

Type of Material	Material characteristics		SA integration method		Laser properties				Years reported [ref.]			
	Fabrication method	Thickness (nm); number of layers (n)			Laser type	Gain medium	λ (nm)	T		f_{rep}	P_{max} (mW)	
MoS ₂	LPE	Few-layer; $\approx$ 9.2 nm;	Fiber facet	Mode-locking		Yb-doped fiber	1054.3	800 ps	6.51 MHz	9.3	2014 ^[101]	
	LPE	Few-layer; <10 nm	MoS ₂ /PVA film	Q-switching		Yb-doped fiber	1030–1070	2.88 μ s	74 kHz	10.5	2014 ^[158]	
	LPE	Few-layer; $\approx$ 3 layers; $\approx$ 1.87 nm;	MoS ₂ /PVA film			Yb-doped fiber;	1060	5.8 μ s	28.9 kHz	0.9	2014 ^[159]	
	Wet-chemistry ^{a)}		Few-layer	microfiber	Mode-locking		Er-doped fiber	1557	3.0 ps	2.5 GHz	5.39	2014 ^[162]
		PLD ^{b)}	Over 30 layers; $\approx$ 20 nm	Quartz slide	Q-switching		Nd: CdVO ₄ crystal	1060	970 ns	732 kHz	227	2014 ^[102]
						Nd:YGG	1420	729 ns	77 kHz	52		
						Tm:Ho:YGG	2100	410 ns	149 kHz	206		
	LPE	MoS ₂ sheets	MoS ₂ /PVA film	Mode-locking		Er-doped fiber	1550	710 fs	12.09 MHz	17.8	2014 ^[160]	
	LPE	Few-layer; $\approx$ 4 nm; $\approx$ 1–3 layers	Microfiber	Mode-locking		Er-doped fiber	1040	656 ps	6.74 MHz	2.37	2014 ^[166]	
	LPE	Few-layer	MoS ₂ solution	Q-switching		Nd:YAG crystal	1060	2.8 μ s	20.8 kHz	23.5	2015 ^[167]	
	LPE	Few-layer; <10 nm	MoS ₂ /PVA film	Mode-locking		Er-doped fiber	1335–1565	0.96 ps–7.1 ps	12.99 MHz	–	2015 ^[161]	
	LPE	Few-layer; <10 nm	MoS ₂ /PVA film	Mode-locking		Er-doped fiber	1560	935 fs	463 MHz	5.9	2015 ^[168]	
	LPE	3–4 layers; 2–3 nm	MoS ₂ /PVA film	Q-switching		Pr-doped ZBLAN fiber	604	602 ns	118.4 kHz	0.6	2016 ^[169]	
Solvothermal		Nanosheets; <5 nm	MoS ₂ /PVA film	Q-switching		Er-doped fiber	1560	1.9 μ s	64.9 kHz	3.1	2016 ^[170]	
	LPE	3–4 layers; 2–3 nm	MoS ₂ /PVA film	Q-switching		Pr-doped ZBLAN fiber	635	227 ns	438.6 kHz	7.1	2016 ^[165]	
	LPE	–	MoSe ₂ /PVA film	Q-switching		Yb-doped fiber;	1060	2.8 μ s	74.9 kHz	8.72	2015 ^[163]	
						Er-doped fiber;	1560	4.8 μ s	35.4 kHz	29.2		
						Tm-doped fiber	1920	5.5 μ s	21.8 kHz	0.9		
MoSe ₂	LPE	–	MoSe ₂ /PVA film	Mode-locking		Er-doped fiber	1560	580 fs	8.8 MHz	0.8	2015 ^[171]	
	LPE	$\approx$ 80 nm	MoSe ₂ /PVA film	Mode-locking		Er-doped fiber	1557.3	751 fs	3.27 GHz	22.8	2016 ^[164]	
	LPE	3–6 layers; 2–4 nm	MoSe ₂ /PVA film	Q-switching		Pr-doped ZBLAN fiber	635	240 ns	555.1 kHz	6.2	2016 ^[165]	
	LPE	3 nm	MoTe ₂ /PVA film	Mode-locking		Er-doped fiber	1557	1 ps	–	–	2016 ^[172]	
			Microfiber	Mode-locking		Er-doped fiber	1561	1.2 ps	5.26 MHz	–		
MoTe ₂	LPE	3–4 layers; 2–3 nm	WS ₂ /PVA film	Q-switching		Pr-doped ZBLAN fiber	604	435 ns	132.2 kHz	0.7	2016 ^[169]	
	PLD	–	Microfiber	Mode-locking		Er-doped fiber	1559.7	452 fs	1.04 GHz	11.3	2015 ^[173]	
WS ₂	LPE	2–20 nm	Microfiber	Mode-locking		Er-doped fiber	1557	1.32 ps	8.86 MHz	110	2015 ^[174]	
		$\approx$ 30 nm	Microfiber	Mode-locking		Er-doped fiber	1558	1.2 ps	8.96 MHz	–		
	LPE		WS ₂ /PVA film	Mode-locking		Er-doped fiber	1559.1	840 fs	14.57 MHz	0.034	2016 ^[175]	

Table 2. Continued.

Type of Material	Material characteristics		SA integration method		Laser properties				Years reported, [ref.]		
	Fabrication method	Thickness (nm); number of layers (n)			Laser type	Gain medium	λ (nm)	T		f_{rep}	P_{max} (mW)
WSe ₂	Commercial nanosheet solution	≈23.1 nm	Photonic crystal fiber	Mode-locking	Er-doped fiber		1563.8	524 fs	19.57 MHz	2.64	2015 ^[176]
	LPE	2–4 nm; 3–6 layers	WS ₂ /PVA film	Q-switching	Pr-doped ZBLAN fiber		635	207 ns	512.8 kHz	8.7	2016 ^[165]
	LPE		WSe ₂ /PVA film	Q-switching	Er-doped fiber		1561	4.063 μs	85.365 kHz	375	2015 ^[177]
	LPE	≈20 nm	Microfiber	Mode-locking	Er-doped fiber		1556.7	1.31 ps	5.31 MHz	0.84	2016 ^[178]
	LPE	2.5 nm	WTe ₂ /PVA film	Mode-locking	Er-doped fiber		1562	≈1 ps	–	–	2016 ^[172]
WTe ₂	LPE	80–150 nm; 100–200 layers	Microfiber	Mode-locking	Er-doped fiber		1556	≈1 ps	–	–	2016 ^[179]

^a)Synthesized by hydrothermal intercalation and following exfoliation process; ^b)PLD: pulse laser deposition.

4.5. 2D-Black-Phosphorous-Enabled Pulse Lasers

Monolayer or few-layer phosphorene demonstrate similar NLO properties as that of 2D TMDs. Nevertheless, the smaller bandgap of phosphorene in few-layer stacks demonstrate much larger bandwidth for saturable absorption compared with TMDs. In addition to application as NIR SAs, phosphorene has shown to be an excellent MIR SA for either mode-locking or Q-switching pulse generation in spectral region of 2 μm and longer. The development of BP-based SAs for ultrafast pulse generation began in 2015, following the report of its exotic electric and optical properties.^[12] As it can be processed by techniques applied for other 2D materials, the BP-based SAs therefore can be easily fabricated using existing mature methods. In the most common 1550 nm band (using Er-doped gain fiber), BP-based SAs have enabled both Q-switching and mode-locking pulse generation in the past 2 years.^[180–184] The first BP-based SA was reported by Chen et al.^[180] using mechanically exfoliated BP few-layer sheets.^[180] The BP sheet was coated on the fiber facet to enable both Q-switching and mode-locking. This work demonstrated both Q-switching pulse with a maximum pulse energy of 94.3 nJ and mode-locking pulse with pulse duration of 946 fs using Er-doped gain fiber. Following this report, mode-locking pulse based on a similar laser system was demonstrated by Sotor et al.^[182] using thick BP sheets as SA (around 300 nm thickness). The output optical solitons at around 1550 nm have a bandwidth of 10.2 nm and a pulse duration of 272 fs. Liquid phase exfoliated BP with much smaller thickness can be also processed into thin film SA by blending with PMMA.^[181] The resultant SA enabled Q-switching pulse generation with a pulse energy of 194 nJ and pulse duration of 4.35 μs using an Er-doped fiber laser. Mode-locking laser pulse was also generated but with a very high threshold (> 100 mW) and high tendency to instability due to the high nonsaturable loss of SA. In a similar fashion, SA made from BP quantum dots derived from a solvothermal process also realized mode-locking pulse output from an Er-doped fiber with pulse duration of 1.08 ps.^[184]

Along with the development of the NIR pulse lasers using BP-based SA, a visible laser operating at 635 nm with a pulse duration of 383 ns was also demonstrated based on a Pr-doped fluoride fiber.^[183] Possibly due to the high nonsaturable loss in the visible region, mode-locking pulse was not achieved in their experiment. MIR mode-locking pulse generation was first demonstrated by Sotor et al.^[185] using 300 nm BP sheets (around 500 layers). This laser employs a Tm-doped fiber and the BP SA was inserted in between the fiber connector. The laser generates 793 fs pulse at the central wavelength of 1910 nm.

The main obstacle to realize high power lasers using these strong absorbing SAs including BP and other 2D materials are the low thermal damage threshold due to heating by the laser. Lee et al.^[186] developed a new strategy for the integration of SA into the laser system by coating the SA materials (BP) onto a side-polished fiber, in which the saturable absorption occurs via the evanescent interaction (**Figure 13**). This laser is based on an Er-doped fiber, generating a 805 fs pulse with repetition rate of 3.82 MHz. In particular, within a period of 168 h, both the repetition rate and the spectral width were unchanged, indicating very stable operation of the laser. This technique could

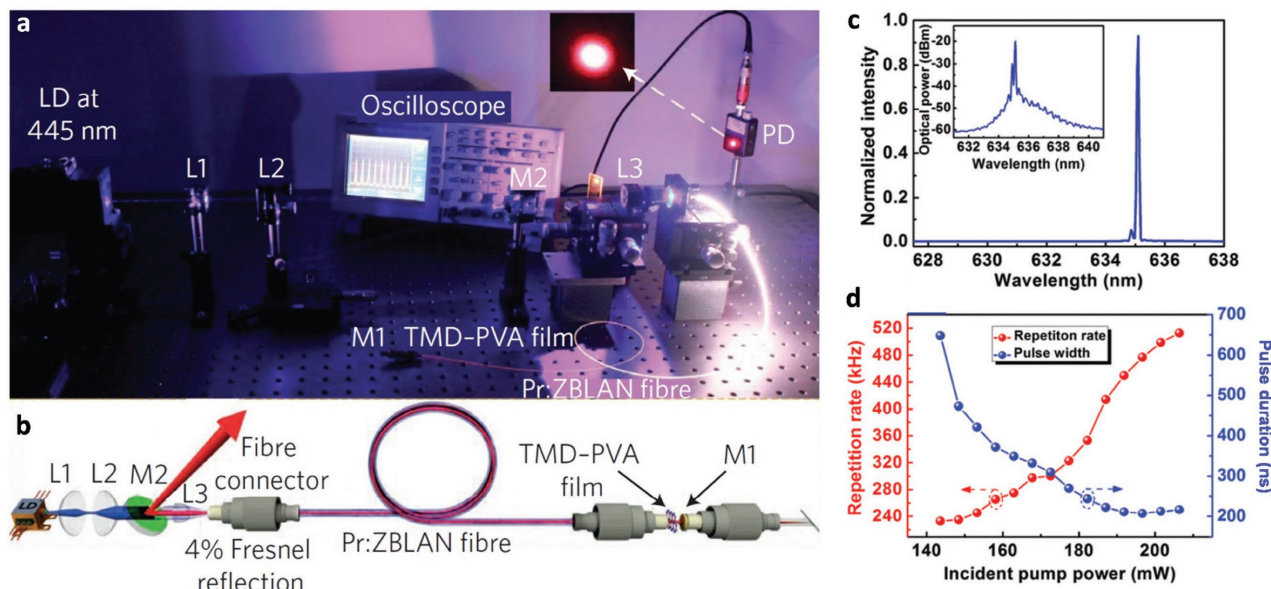


Figure 12. A Q-switched visible laser enabled by TMD-based SA using Pr-doped fluoride (ZBLAN) fiber. a) Photograph of the visible pulse laser in operation. The inset is the beam spot from the output end. b) Schematic of the experimental setup for the laser. L1–L3 are lenses; M1, M2, mirrors, PD, photodetector; LD, laser diode. c) Q-switched laser output spectrum enable by WS₂-based SA. Inset shows the spectrum in the narrow spectral region. d) Dependence of repetition rate and pulse duration on the incident pump power. Reproduced with permission.^[165] Copyright 2016, the Royal Society of Chemistry.

be exploited further to develop fiber laser operating at much higher power based on other gain fibers.

Besides fiber lasers, solid state laser using BP-based SA was reported by Zhang et al.^[113] Benefiting from the ultra-broad

band saturable absorption, Q-switching pulse lasers at 639 nm, 1060 nm, and 2100 nm were demonstrated successfully based on Pr:LiGdF₄, Nd:GdVO₄ and Tm:Ho:YGG crystals, respectively. Again, because of the high nonsaturable loss

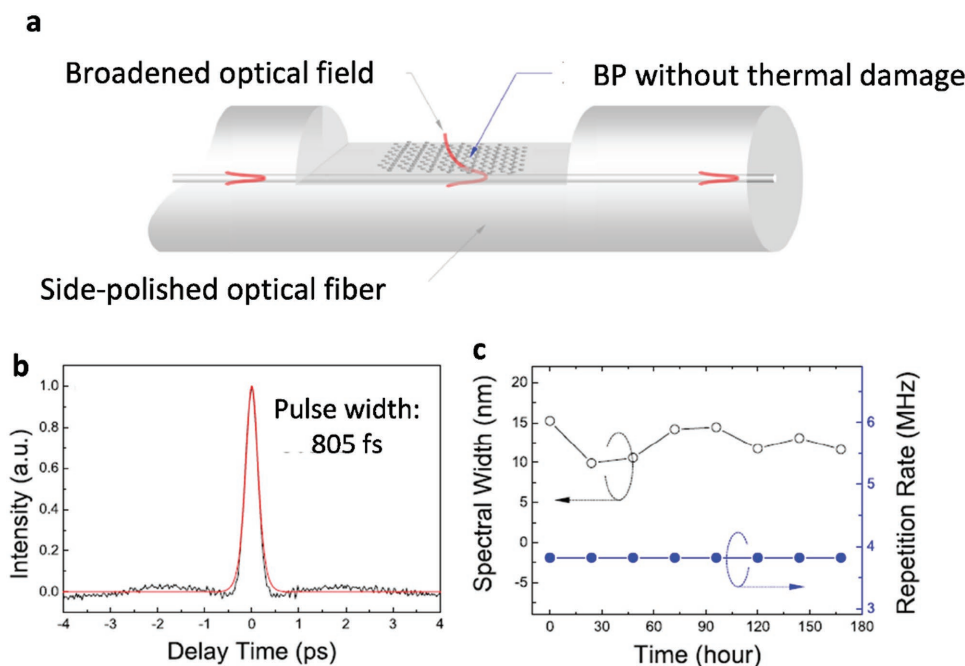


Figure 13. Suppression of thermal damage in a BP-based SA for mode-locking. a) Schematic illustration of the SA by depositing BP on a side-polished fiber. b) Autocorrelation profile of the pulse, showing a pulse duration of 805 fs. The black and the red curves are the experimental and fitting results, respectively. c) The stability of the laser tested by monitoring the spectral width and repetition rate in a timespan of 180 h. Reproduced with permission.^[186] Copyright 2016, IOP Publishing.

mode-locking pulse was not demonstrated. The operation wavelength was extended to 2700 nm in a Q-switching laser reported by Kong et al.^[187] using Er:Y₂O₃ transparent ceramic as the gain medium. The SA was made by deposition of liquid phase exfoliated few-layer BP onto a gold film mirror (with a reflectivity of 90%). Q-switching operation at 2700 nm was realized, with an output power of 6 mW and pulse duration of 4.47 μs. More technical details of these BP-enabled lasers are provided in Table 3.

4.6. Zero-Gap-Materials-Enabled Pulse Lasers

TIs that have been exploited for photonic applications include several group-V heavy metal chalcogenides, such as Bi₂Se₃, Bi₂Te₃ and Sb₂Te₃. Different SAs have been fabricated from TIs preferably in their nanoscale form accessed by either mechanical/liquid phase exfoliation or liquid phase synthesis. The first TI-based SA was reported by Zhao et al.^[116] which enabled a 1.21 ps mode-locking pulse laser at 1558 nm based on an Er-doped fiber. This Bi₂Te₃-based SA with a modulation depth up to 95% was fabricated by coating liquid phase synthesized Bi₂Te₃ onto a quartz plate, which was inserted into the cavity as a free space SA. This laser can also output tunable solitons between 1557 nm and 1565 nm with pulse duration of 1.57 ps.^[117] Later on, a number of reports followed, rapidly leading to the demonstration of harmonic mode-locking lasers using SAs based on other TIs, like Bi₂Se₃,^[117] Bi₂Te₃,^[116,119,195] and Sb₂Te₃,^[196] and their alloy Bi₂TeSe₂^[118] using techniques shown in Figure 8. Benefiting from the gapless character of the TIs, pulse laser at the 2 μm region was demonstrated by Yin et al.^[197] based on the evanescent field interaction method. This laser based on a Tm-doped fiber produced a mode-locked pulse at 1909.5 nm with a repetition rate of 21.5 MHz and pulse duration of 1.26 ps. Up to the present, the highest repetition rate enabled by TI-SAs can reach over 2 GHz,^[198] and the shortest pulse generated by TI-SAs is 128 fs based on an Er-doped fiber laser using Sb₂Te₃ SA by an evanescent field interaction technique.^[196] Beside fiber lasers, a solid state pulse laser based on TI-SA has also been developed. In a recent report by Yu et al.,^[120] the SA was fabricated by deposition of Bi₂Se₃ NPs synthesized by wet chemistry onto a quartz plate. This laser based on a Nd:GdVO₄ crystal operated at 1063 nm in a Q-switching mode, producing a maximum output power of 32 mW and pulse energy of 60 nJ (pump power 1.85 W). Table 4 lists the performances of the pulse lasers developed using TI-based SAs.

Similar to the gapless character of TIs, a thin film of the Dirac semimetal Ca₃As₂, known as 3D graphene, has also been fabricated as an SA for the demonstration of a mode-locked pulse laser in the 3 μm MIR region.^[121] This laser is based on a Ho³⁺/Pr³⁺ codoped fluoride fiber and the SA was fabricated by deposition of the Cd₃As₂ film onto a gold protected mirror (Figure 14). Therefore, the SA operates in reflection. At a pumping power at 80 mW, the system changes from Q-switched mode-locking to CW (constant-wave) mode-locking. For CW mode-locking, this laser operates at 2860 nm with a pulse width of 6.2 nm and pulse duration of 9.8 ps. Provided with suitable RE-doped fluoride or chalcogenide fibers, the pulse lasers operating in

wavelength over 3 μm are expected to be produced based on the SAs made from gapless materials like TIs and semimetals.

4.7. SAs Based on Hybrid Systems

As 2D materials are separated or stacked few-layers, they can be engineered to build van der Waals heterostructures, offer surprising new opportunities for novel devices.^[212] Such hybrid systems can be made from graphene, 2D semiconductors, and the quasi-2D TIs. Beside electric and optical properties, recently such hybrid materials have also been tested for saturable absorption and short pulse generation.^[213–215] For instance, graphene–Bi₂Te₃ van der Waals heterostructures were made by Mu et al.^[214] by overgrowth of a layer of Bi₂Te₃ onto the graphene layer using a CVD process. It was found that the carrier relaxation time and modulation depth both increase with the coverage of Bi₂Te₃ on graphene. The increased modulation depth can be easily understood by considering the enhanced absorption for thicker films, while the slowing down of carrier cooling was explained by the additional collisions of carriers from the Bi₂Te₃ layer. Based on an Er-doped fiber laser, the author successfully demonstrated a Q-switching laser and a mode-locking laser at 1550 nm. For the Q-switching laser, the maximum output power reached 3.2 mW with a pulse energy of 93 nJ. Mode-locking started at a threshold power of 40 mW. Under a pumping power of 140.7 mW, the soliton pulse has a pulse duration of 837 fs with a single pulse energy 0.178 nJ (17.3 MHz). A similar graphene–Bi₂Te₃ hybrid film was used as SA for the generation of mode-locking pulse at both the 1550 nm and 1000 nm regions based on Er, and Yb-doped fibers, respectively.^[215] Compared with the single-component SAs, the hybrid systems offer an obvious advantage; that is, materials can be designed by tuning the composition, coverage of the structures, to obtain SAs with tailored performance.

4.8. General Considerations for Selection of a Suitable SA based on New LD Materials

Having analyzed the recent progress on different SAs made from graphene, 2D semiconductors and zero-gap materials of TIs and semimetals, we are now able to give a few balanced guidelines (or may be considered biased) concerning the selection of SAs that meets the conditions required by specific laser applications, by taking into account of their quite different characters, such as responsive bandwidth, time, modulation depth, etc. Despite that laser output is determined by not only the performance of the SA used but also the laser system itself (i.e., gain medium), several general guidelines still can be suggested that indeed related to the characters of the SAs discussed above. First, the presence of an optical bandgap still determines the working wavelength of SAs. This is why zero gap materials (e.g., graphene, TIs) have enabled MIR laser pulse generation that cannot be easily accessed by most TMD-based SAs. Whereas apparently a same laser working at the same wavelength (e.g., 1550 nm band) can be made with SAs based on quite different LD materials which demonstrate rather different NLO properties, a critical question has to be answered: which one is the

Table 3. BP-based SA enabled pulse lasers.

Material characteristics		SA integration method	Laser properties		Years reported [ref.]
Fabrication method	Thickness (nm); number of layers (n)		Gain medium	λ (nm)	
ME	15–25 layers	Fiber facet	Mode-locking	1570 nm	2015 ^[180]
ME	≈300 nm	Fiber facet	Mode-locking	1560.5	2015 ^[182]
ME	Few-layer; 25–80 nm	BP film @ quartz wafer	Q-switching	636	2015 ^[113]
LPE	–	Fiber facet	Mode-locking	1060	2015 ^[188]
LPE	4–25 nm	BP/pVA nanofiber film	Q-switching	1565.9	2015 ^[181]
LPE	5–20 nm	SA mirror	Q-switching	2779	2015 ^[189]
ME	200–300 nm	Deposited on fiber connector	Mode-locking	1910	2015 ^[185]
LPE	0.6–2 nm; 1–3 layers	Microfiber	Mode-locking	1566.5	2015 ^[190]
LPE	≈5 nm; 8 layers	SA mirror	Mode-locking	1064.1	2015 ^[191]
Solvothermal method	QDs; 2.1 ± 0.9 nm	BPQDs/PVDF film	Mode-locking	1567.5	2016 ^[184]
ME	≈40 nm	BP @ side polished fiber	Mode-locking	1559	2016 ^[186]
ME	Few-layer; 22 nm	BP @ gold film mirror	Q-switching	1030	2016 ^[187]
ME	–	SA mirror	Q-switching	1930	2016 ^[192]
LPE	50–100 nm	SA mirror	Mode-locking	2009	2016 ^[193]
LPE	≈2.8 nm; ≈3 layers	BP/pVA	Q-switching	635	2016 ^[193]
ME	≈143 nm; 238 layers	SA mirror	Mode-locking	2783	2016 ^[193]

Table 4. Other (TIs and Dirac-semimetal) based SA enabled pulse lasers.

Type of Material	Material characteristics		SA integration method	Laser properties				Years reported [ref.]
	Fabrication method	Thickness (nm); number of layers (n)		Laser type	Gain medium	λ (nm)	T	
Bi_2Se_3	Polyol method	–	Quartz slide	Mode-locking	Er-doped fiber	1564.6	1.57 ps	2012 ^[117]
	Polyol method	–	Fiber facet	Q-switching	Er-doped fiber	1565.1	13.4 μs	2013 ^[199]
	Polyol method	–	Quartz slide	Q-switching	Nd:GdVO ₄ crystal	1060	666 ns	2013 ^[120]
Bi_2Te_3	Polyol method	–	Bi_2Se_3 /PVA	Mode-locking	Er-doped fiber	1557.5	660 fs	2014 ^[200]
	LPE	3–4 nm; 4 layers	Quartz slide	Q-switching	Er-doped fiber	1530.3	4.9 μs	2014 ^[201]
	LPE	–	Quartz slide	Mode-locking	Er-doped fiber	1558.4	1.21 ps	2012 ^[116]
	LPE	–	Microfiber	Mode-locking	Er-doped fiber	1558.5	2.49 ps	2013 ^[198]
	LPE	≈ 18.5 nm	Quartz slide	Q-switching	Er:YAG ceramic	1645	6.03 μs	2013 ^[202]
	LPE	–	SA mirror	Q-switching	Nd:YVO ₄ crystal	1064	≈ 1 μs	2014 ^[203]
	LPE	–	Microfiber	Q-switching	Er-doped fiber	1543.2	9.5 μs	2014 ^[204]
	ME	≈ 15 μm	Microfiber	Mode-locking	Er-doped fiber	1547	≈ 600 fs	2014 ^[205]
Sb_2Te_3	LPE	≈ 20 nm	Bi_2Te_3 NS solution filled in PCF	Mode-locking	Yb-doped fiber	1060.47	960 ps	2015 ^[206]
	hydrothermal	20 nm	Deposited on tapered fiber	Mode-locking	Tm, Ho-codoped fiber	1909.5	1.26 ps	2015 ^[197]
	LPE	7–11 nm	Bi_2Te_3 on CaF_2 substrate	Q-switching	Ho-doped ZBLAN fiber	2979.9	1.37 μs	2015 ^[207]
Sb_2Te_3	Polyol method	≈ 10 nm	Bi_2Te_3 /PMMA film	Q-switching	Er-doped fiber	1531	17 μs	2015 ^[119]
	ME	40–100 nm	Fiber facet	Mode-locking	Er-doped fiber	1558.6	1.8 ps	2014 ^[208]
	Bulk sample	0.5 mm	Side-polished fiber	Mode-locking	Er-doped fiber	1565	128 fs	2014 ^[196]
Bi_2SeTe_2	ME	–	Microfiber	Mode-locking	Er-doped fiber	1560	270 fs	2014 ^[209]
	ME	–	Fiber facet	Mode-locking	Tm-doped fiber	1945	890 fs	2016 ^[210]
	Magnetron sputtering	–	Microfiber	Mode-locking	Yb-doped fiber	1047.1	5.9 ps	2016 ^[211]
Bi_2SeTe_2	polyol	–	BST/PMMA film inserted into fiber connectors	Q-switching	Er-doped fiber	1560	16.4 μs	2015 ^[118]
Cd_3As_2	MBE	–	SA mirror	Q-switched mode-locking	$\text{Ho}^{3+}/\text{Pr}^{3+}$ co-doped fluoride fiber	2860	6.3 ps	2016 ^[121]

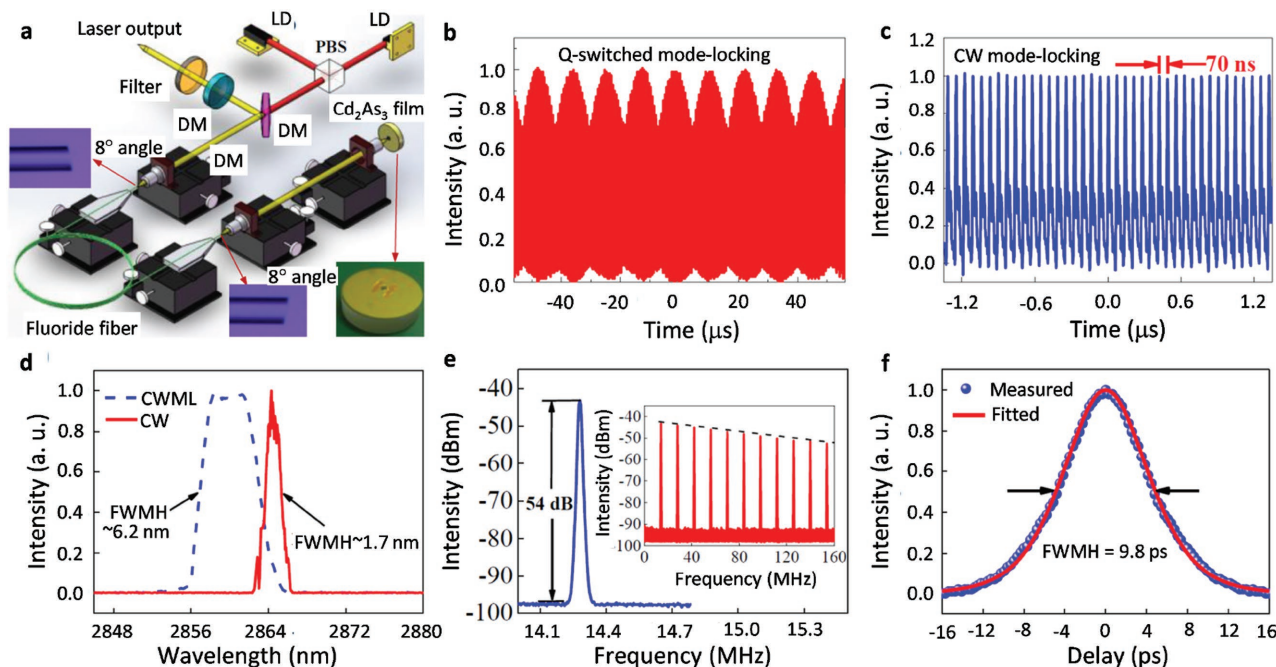


Figure 14. MIR pulse laser operating at 2.8 μm using a Cd₃As₂ SA. a) Schematic of the experimental setup for the fiber laser system. b) Q-switched mode-locking pulse train obtained at a pumping power of 57.6 mW. c) CW mode-locking pulse train obtained at a pumping power of 286.9 mW. d) Laser spectrum of the CW mode-locking (CWML) pulse and the CW pulse. e) RF spectrum recorded at the repetition rate of 14.28 MHz (with the resolution of 10 kHz). The signal-to-noise ratio is 54 dB. f) Autocorrelation profile of pulse and the fitting by Sech² function (red curve). The FWHM is 9.8 ps. Reproduced with permission.^[121] Copyright 2017, Nature Publishing Group.

best? As the second guideline, a natural choice from practical considerations would be the SA with best energy efficiency (i.e., output/input ratio) that is associated with non-saturable loss, which depends on intrinsic properties (like absorption coefficient) as well as extrinsic properties (such as scattering loss). Since heat accumulation due to non-saturable loss is unavoidable for the SAs, LD materials with the best thermal and chemical robustness are therefore suggested for making SAs. Unfortunately, the thermal stability as well as oxidation resistance of chalcogenide compounds as well as BP have strongly limited the maximum output power and plagued the long term stability of developed laser systems.^[186] Therefore in the third guideline, we suggest, based on our experimental results, the selection of SA materials by referring on the following stability sequence: oxide plasmonic NCs > graphene, 2D metal sulfide TMDs > BP and other metal compounds (selenide, telluride, etc.). Beside output power and stability, the pulse duration and repetition rate are also affected by the type of properties of SA, such as response time. These parameters are nevertheless more closely linked with characteristics of the gain medium and laser cavity parameter, which, on the other hand, should be designed to adapt to these new SAs.

5. Other Nonlinear Optical Devices based on LD Materials

The above LD materials have demonstrated excellent nonlinear optical properties and ultrafast response. The NLO absorption manifests high third order susceptibility as well as Kerr

susceptibility. Besides SA for ultrafast laser systems, these LD materials have found important applications associated with their nonlinear optical properties. While SA makes use of the optical loss of the materials manifested by the imaginary part of the complex susceptibility, materials with the high real part of the susceptibility, the nonlinear refractive index, can be explored for all-optical control and a number of photonic applications. Historically, materials with high $\chi^{(3)}$, such as metal NPs and semiconductor QDs, have been used for the demonstration of ultrafast optical switch in the visible and NIR region.^[4,186,216–218]

Recently, by making use of the nonlinear optical Kerr effect, graphene and TI have been used for all-optical signal processing and optical modulation. This means the control of intensity or the propagation of light with another controlling light field. For instance, an optical Kerr switcher was demonstrated by Chen et al. by using a TI-coated microfiber based on the evanescent field interaction.^[219] The optical Kerr switcher is an important component in all-optical circuits and networks. In their experiment, an Er-doped fiber is used as the active medium and a pump signal with sufficient high intensity was used to rotate the polarization of the probe signal based on the optical Kerr effect arising from the large nonlinear refractive index of TI. Therefore, the probe signal can be modulated to exhibit an on/off state by switching on/off the pump signal. Based on the same TI-coated microfiber, the authors also demonstrate a wavelength converter based on the four-wave mixing technique, producing a maximum conversion efficiency of −34 dB.

A similar device for all optical modulation is reported by Yu et al.^[220] using graphene-coated microfiber based on the optical Kerr phase shift. By making use of the evanescent

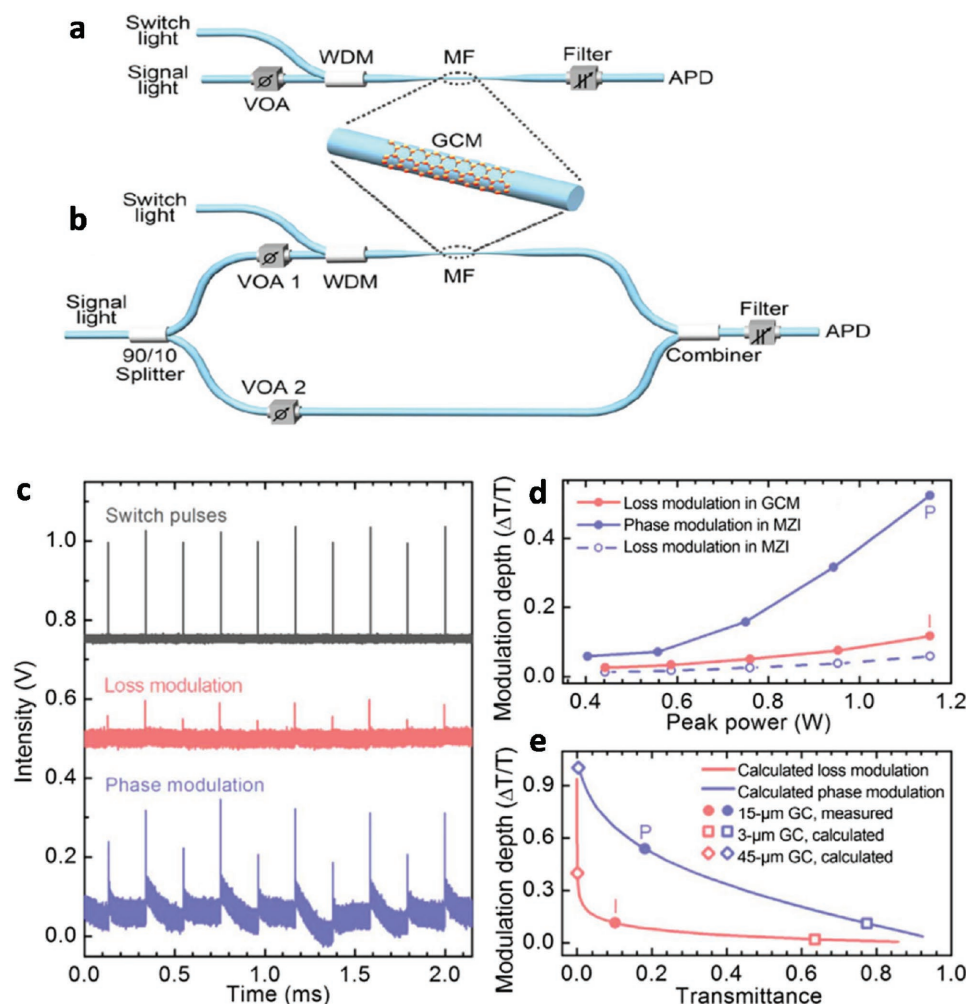


Figure 15. All-optical modulation based on optical Kerr effect of graphene. a, b) Schematics of the device structures using graphene-clad microfiber (GCM) for all-optical modulation: a) optically induced loss modulation, and b) phase modulation using Mach–Zehnder interferometer (MZI). APD, avalanche photodetector; VOA, variable optical attenuator; MF, microfiber; WDM, wavelength division multiplexing; c) Loss (middle) and phase (bottom) modulated pulse train by the switching pulse (top). d) Modulation depth as a function of peak power (switch signal) for loss and phase modulation. e) Calculated and observed modulation depth as a function of transmittance. Reproduced with permission.^[220] Copyright 2016, Optical Society of America.

field interaction, they have realized both optically induced loss modulation and phase modulation with the graphene-clad microfiber based on the Mach–Zehnder interferometer (MZI) structure (Figure 15). In their devices, a pulse laser operating at 1064 nm (8 ns, 4.8 kHz) is used as the switching light to induce the optical Kerr effect, such that the optical loss (Figure 15a) and phase (Figure 15b) both can be modulated. The results showed that modulation depth (defined as $\Delta T/T$) increases with the switching signal power and it reaches a maximum value of around 0.5 for phase modulation by MZI (Figure 15). In addition, it is shown clearly that a lower transmittance favors a higher modulation depth for both loss and phase modulation using the graphene-clad microfiber.

6. Conclusions and Perspectives

The past decade has witnessed the booming of investigations into dozens of novel materials, including graphene, different

2D semiconductors, quasi-2D TIs and semimetal materials, which have been extensively explored for diverse potential applications. Besides the intriguing electric and optical properties, we have discussed here that they also exhibit excellent nonlinear optical response, as manifested by the strong saturable absorption (or reverse saturable absorption). Combined with the facile accessibility usually by the liquid phase route (or mechanical exfoliation) and easy device integration, these emerging materials, preferably in their low dimensional form, have showed great promise for applications in nonlinear optics and ultrafast photonics. In particular, they have enabled short pulse laser generation based on both Q-switching and mode-locking in bulk and fiber lasers. All the SAs based on these materials reported so far have demonstrated a working bandwidth far larger than that of commercial SESAMs. Benefited from the large saturable absorption bandwidth and ultrafast carrier cooling rate, laser operating at wavelength region from visible to the MIR region have been developed, with pulse duration down to 30-fs scale. Nevertheless, SAs fabricated from

different LD materials certainly bear clear differences in the modulation depth, saturation intensity, relaxation time and etc., depending on not only the intrinsic material properties (such as bandgap) but also extrinsic properties and fabrication process.

Admittedly, one of the critical issues in the commercialization of the discussed SAs probably lies in the long term stability, especially under high power operation. Clever laser designs can partly circumvent this problem; while for practical application SAs, made from chalcogenides and phosphorous, as well as the polymers (i.e., PMMA, Poly(methyl methacrylate), as matrix) should be carefully protected against oxidation and thermal damage. Up to the present, lasing performance in most proof-of-concept demonstrations are still moderate and Watt-level fiber lasers operating in the IR and MIR range by using SA based on these LD materials still have not been reported. In addition, to fully exploit these LD materials based SA for pulse lasers, the NLO parameters including modulation depth, non-saturable loss and recovery rate should be carefully adapted to suppress Q-switching instabilities and then to achieve higher repetition rate and output power.

Besides the discussed materials, new possible LD materials for the development of better SAs might be realized by combination of two or more single materials, where the performance can be further enhanced due to a synergic effect. For instance, in a van der Waals heterojunction made by stacking different 2D materials that allow for tailored optical properties by choosing suitable 2D components,^[212,221] the NLO can be optimized with respect to the saturable absorption bandwidth as well as the response speed. In addition to 2D materials, semiconductor plasmonic materials based on refractory oxides or nitrides are especially robust against thermal damage in addition to the broadband NLO response;^[222] therefore they could be less fancy, yet more realistic choices suitable for high power, high energy operation. Along with material optimization and nanoscale structuring, first principle calculation based on density functional theory can correlate optical absorption with the electronic structure and provide useful information to aid the design of new SAs. Finally, in order to transfer the lab-scale fabrication to mass production for commercial use, new techniques for fabrication of SAs have to be applied, including commercially viable thin film processing methods such as vapor phase growth and deposition. It is expected that in the near future these new SAs would prevail over conventional costly SESAMs with only short working bandwidth, and the current high record of pulse duration, average power and pulse repetition rate can be pushed further by making use of the new SAs.

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- [1] V. M. Axt, S. Mukamel, *Rev. Mod. Phys.* **1998**, *70*, 145.
- [2] L. W. Tutt, T. F. Boggess, *Prog. Quant. Electron.* **1993**, *17*, 299.
- [3] J. Wang, W. J. Blau, *J. Opt. A-Pure Appl. Op.* **2009**, *11*, 024001.
- [4] A. L. Stepanov, *Rev. Adv. Mater. Sci.* **2011**, *27*, 115.
- [5] L. W. Tutt, A. Kost, *Nature* **1992**, *356*, 225.
- [6] P. Chen, X. Wu, X. Sun, J. Lin, W. Ji, K. L. Tan, *Phys. Rev. Lett.* **1999**, *82*, 2548.
- [7] A. Martinez, Z. P. Sun, *Nat. Photonics* **2013**, *7*, 842.
- [8] S. Yamashita, *J. Lightwave. Technol.* **2012**, *30*, 427.
- [9] M. S. Xu, T. Liang, M. M. Shi, H. Z. Chen, *Chem. Rev.* **2013**, *113*, 3766.
- [10] K. F. Mak, J. Shan, *Nat. Photonics* **2016**, *10*, 216.
- [11] Q. H. Wang, K. Kalantar-Zadeh, A. Kis, J. N. Coleman, M. S. Strano, *Nat. Nanotechnol.* **2012**, *7*, 699.
- [12] X. Ling, H. Wang, S. X. Huang, F. N. Xia, M. S. Dresselhaus, *P. Natl. Acad. Sci. USA* **2015**, *112*, 4523.
- [13] D. Dini, M. J. F. Calvete, M. Hanack, *Chem. Rev.* **2016**, *116*, 13043.
- [14] Y. Chen, T. Bai, N. N. Dong, F. Fan, S. F. Zhang, X. D. Zhuang, J. Sun, B. Zhang, X. Y. Zhang, J. Wang, W. J. Blau, *Prog. Mater. Sci.* **2016**, *84*, 118.
- [15] A. P. Alivisatos, *Science* **1996**, *271*, 933.
- [16] L. E. Brus, *J. Chem. Phys.* **1984**, *80*, 4403.
- [17] R. C. Ashoori, *Nature* **1996**, *379*, 413.
- [18] A. D. Yoffe, *Adv. Phys.* **2001**, *50*, 1.
- [19] A. D. Yoffe, *Adv. Phys.* **1993**, *42*, 173.
- [20] E. H. Sargent, *Adv. Mater.* **2005**, *17*, 515.
- [21] S. A. McDonald, G. Konstantatos, S. G. Zhang, P. W. Cyr, E. J. D. Klem, L. Levina, E. H. Sargent, *Nat. Mater.* **2005**, *4*, 138.
- [22] T. Itoh, Y. Iwabuchi, M. Kataoka, *Phys. Status Solidi B* **1988**, *145*, 567.
- [23] P. K. Jain, K. S. Lee, I. H. El-Sayed, M. A. El-Sayed, *J. Phys. Chem. B* **2006**, *110*, 7238.
- [24] C. Noguez, *J. Phys. Chem. C* **2007**, *111*, 3806.
- [25] K. L. Kelly, E. Coronado, L. L. Zhao, G. C. Schatz, *J. Phys. Chem. B* **2003**, *107*, 668.
- [26] K. M. Mayer, J. H. Hafner, *Chem. Rev.* **2011**, *111*, 3828.
- [27] S. Szunerits, R. Boukherroub, *Chem. Commun.* **2012**, *48*, 8999.
- [28] X. M. Qian, S. M. Nie, *Chem. Soc. Rev.* **2008**, *37*, 912.
- [29] J. M. Luther, P. K. Jain, T. Ewers, A. P. Alivisatos, *Nat. Mater.* **2011**, *10*, 361.
- [30] G. V. Naik, V. M. Shalae, A. Boltasseva, *Adv. Mater.* **2013**, *25*, 3264.
- [31] A. Comin, L. Manna, *Chem. Soc. Rev.* **2014**, *43*, 3957.
- [32] X. Liu, M. T. Swihart, *Chem. Soc. Rev.* **2014**, *43*, 3908.
- [33] K. S. Novoselov, A. K. Geim, S. V. Morozov, D. Jiang, Y. Zhang, S. V. Dubonos, I. V. Grigorieva, A. A. Firsov, *Science* **2004**, *306*, 666.
- [34] R. R. Nair, P. Blake, A. N. Grigorenko, K. S. Novoselov, T. J. Booth, T. Stauber, N. M. R. Peres, A. K. Geim, *Science* **2008**, *320*, 1308.
- [35] J. N. Coleman, M. Lotya, A. O'Neill, S. D. Bergin, P. J. King, U. Khan, K. Young, A. Gaucher, S. De, R. J. Smith, I. V. Shvets, S. K. Arora, G. Stanton, H. Y. Kim, K. Lee, G. T. Kim, G. S. Duesberg, T. Hallam, J. J. Boland, J. J. Wang, J. F. Donegan, J. C. Grunlan, G. Moriarty, A. Shmeliov, R. J. Nicholls, J. M. Perkins, E. M. Grieveson, K. Theuvsen, D. W. McComb, P. D. Nellist, V. Nicolosi, *Science* **2011**, *331*, 568.
- [36] V. Nicolosi, M. Chhowalla, M. G. Kanatzidis, M. S. Strano, J. N. Coleman, *Science* **2013**, *340*, 1420.
- [37] A. C. Ferrari, F. Bonaccorso, V. Fal'ko, K. S. Novoselov, S. Roche, P. Boggild, S. Borini, F. H. L. Koppens, V. Palermo, N. Pugno, J. A. Garrido, R. Sordan, A. Bianco, L. Ballerini, M. Prato, E. Lidorikis, J. Kivioja, C. Marinelli, T. Ryhanen, A. Morpurgo, J. N. Coleman, V. Nicolosi, L. Colombo, A. Fert, M. Garcia-Hernandez, A. Bachtold, G. F. Schneider, F. Guinea, C. Dekker, M. Barbone, Z. P. Sun, C. Galiotis, A. N. Grigorenko,

- G. Konstantatos, A. Kis, M. Katsnelson, L. Vandersypen, A. Loiseau, V. Morandi, D. Neumaier, E. Treossi, V. Pellegrini, M. Polini, A. Tredicucci, G. M. Williams, B. H. Hong, J. H. Ahn, J. M. Kim, H. Zirath, B. J. van Wees, H. van der Zant, L. Occhipinti, A. Di Matteo, I. A. Kinloch, T. Seyller, E. Quesnel, X. L. Feng, K. Teo, N. Rupasinghe, P. Hakonen, S. R. T. Neil, Q. Tannock, T. Lofwander, J. Kinaret, *Nanoscale* **2015**, *7*, 4598.
- [38] K. F. Mak, C. Lee, J. Hone, J. Shan, T. F. Heinz, *Phys. Rev. Lett.* **2010**, *105*, 136805.
- [39] A. Splendiani, L. Sun, Y. B. Zhang, T. S. Li, J. Kim, C. Y. Chim, G. Galli, F. Wang, *Nano Lett.* **2010**, *10*, 1271.
- [40] G. Eda, H. Yamaguchi, D. Voiry, T. Fujita, M. W. Chen, M. Chhowalla, *Nano Lett.* **2011**, *11*, 5111.
- [41] K. F. Mak, K. L. He, J. Shan, T. F. Heinz, *Nat. Nanotechnol.* **2012**, *7*, 494.
- [42] T. Cao, G. Wang, W. P. Han, H. Q. Ye, C. R. Zhu, J. R. Shi, Q. Niu, P. H. Tan, E. Wang, B. L. Liu, J. Feng, *Nat. Commun.* **2012**, *3*, 5.
- [43] L. K. Li, Y. J. Yu, G. J. Ye, Q. Q. Ge, X. D. Ou, H. Wu, D. L. Feng, X. H. Chen, Y. B. Zhang, *Nat. Nanotechnol.* **2014**, *9*, 372.
- [44] S. Zhang, J. Yang, R. J. Xu, F. Wang, W. F. Li, M. Ghufuran, Y. W. Zhang, Z. F. Yu, G. Zhang, Q. H. Qin, Y. R. Lu, *ACS Nano* **2014**, *8*, 9590.
- [45] X. M. Wang, A. M. Jones, K. L. Seyler, V. Tran, Y. C. Jia, H. Zhao, H. Wang, L. Yang, X. D. Xu, F. N. Xia, *Nat. Nanotechnol.* **2015**, *10*, 517.
- [46] C. L. Kane, E. J. Mele, *Phys. Rev. Lett.* **2005**, *95*, 226801.
- [47] M. Konig, S. Wiedmann, C. Brune, A. Roth, H. Buhmann, L. W. Molenkamp, X. L. Qi, S. C. Zhang, *Science* **2007**, *318*, 766.
- [48] L. Muechler, H. J. Zhang, S. Chadov, B. H. Yan, F. Casper, J. Kubler, S. C. Zhang, C. Felser, *Angew. Chem. Int. Ed.* **2012**, *51*, 7221.
- [49] H. L. Peng, W. H. Dang, J. Cao, Y. L. Chen, W. Wu, W. S. Zheng, H. Li, Z. X. Shen, Z. F. Liu, *Nat. Chem.* **2012**, *4*, 281.
- [50] B. L. Yao, L. Y. Ren, X. Hou, *J. Opt. Soc. Am. B* **2003**, *20*, 1290.
- [51] G. Sobon, *Photon. Res.* **2015**, *3*, A56.
- [52] V. E. Babicheva, A. Boltasseva, A. V. Lavrinenko, *Nanophotonics* **2015**, *4*, 165.
- [53] A. D. Neira, N. Olivier, M. E. Nasir, W. Dickson, G. A. Wurtz, A. V. Zayats, *Nat. Commun.* **2015**, *6*, 7757.
- [54] P. K. Jain, K. Manthiram, J. H. Engel, S. L. White, J. A. Fauchaux, A. P. Alivisatos, *Angew. Chem. Int. Ed.* **2013**, *52*, 13671.
- [55] T. S. Luk, D. de Ceglia, S. Liu, G. A. Keeler, R. P. Prasankumar, M. A. Vincenti, M. Scalora, M. B. Sinclair, S. Campione, *App. Phys. Lett.* **2015**, *106*, 151103.
- [56] N. Kinsey, C. DeVault, J. Kim, M. Ferrera, V. M. Shalae, A. Boltasseva, *Optica* **2015**, *2*, 616.
- [57] S. Kalusniak, S. Sadofev, F. Henneberger, *Phys. Rev. Lett.* **2014**, *112*, 137401.
- [58] M. Z. Alam, I. De Leon, R. W. Boyd, *Science* **2016**, *352*, 795.
- [59] L. Caspani, R. P. M. Kaipurath, M. Clerici, M. Ferrera, T. Roger, J. Kim, N. Kinsey, M. Pietrzyk, A. Di Falco, V. M. Shalae, A. Boltasseva, D. Faccio, *Phys. Rev. Lett.* **2016**, *116*, 233901.
- [60] F. F. Wang, Q. Li, L. Lin, H. L. Peng, Z. F. Liu, D. S. Xu, *J. Am. Chem. Soc.* **2015**, *137*, 12006.
- [61] X. G. Ding, C. H. Liow, M. X. Zhang, R. J. Huang, C. Y. Li, H. Shen, M. Y. Liu, Y. Zou, N. Gao, Z. J. Zhang, Y. G. Li, Q. B. Wang, S. Z. Li, J. Jiang, *J. Am. Chem. Soc.* **2014**, *136*, 15684.
- [62] J. S. Niezgoda, E. Yap, J. D. Keene, J. R. McBride, S. J. Rosenthal, *Nano Lett.* **2014**, *14*, 3262.
- [63] W. H. Li, R. Zamani, P. R. Gil, B. Pelaz, M. Ibanez, D. Cadavid, A. Shavel, R. A. Alvarez-Puebla, W. J. Parak, J. Arbiol, A. Cabot, *J. Am. Chem. Soc.* **2013**, *135*, 7098.
- [64] Q. B. Guo, Y. H. Yao, Z. H. Luo, Z. P. Qin, G. Q. Xie, M. Liu, J. Kang, S. A. Zhang, G. Bi, X. F. Liu, J. R. Qiu, *ACS Nano* **2016**, *10*, 9463.
- [65] Q. B. Guo, X. F. Liu, J. R. Qiu, unpublished.
- [66] F. Scotognella, G. Della Valle, A. R. S. Kandada, D. Dorfs, M. Zavelani-Rossi, M. Conforti, K. Miszta, A. Comin, K. Korobcheyskaya, G. Lanzani, L. Manna, F. Tassone, *Nano Lett.* **2011**, *11*, 4711.
- [67] G. Della Valle, F. Scotognella, A. R. S. Kandada, M. Zavelani-Rossi, H. B. Li, M. Conforti, S. Longhi, L. Manna, G. Lanzani, F. Tassone, *J. Phys. Chem. Lett.* **2013**, *4*, 3337.
- [68] T. Limmer, J. Feldmann, E. Da Como, *Phys. Rev. Lett.* **2013**, *110*, 217406.
- [69] E. Garmire, *IEEE J. Sel. Top. Quant.* **2000**, *6*, 1094.
- [70] Q. Q. Wang, J. B. Han, H. M. Gong, D. J. Chen, X. J. Zhao, J. Y. Feng, J. J. Ren, *Adv. Funct. Mater.* **2006**, *16*, 2405.
- [71] H. Zhang, Z. L. Hu, Z. J. Ma, M. Gecevicius, G. P. Dong, S. F. Zhou, J. R. Qiu, *ACS Appl. Mater. Interfaces* **2016**, *8*, 2048.
- [72] K. Mansour, M. J. Soileau, E. W. Vanstryland, *J. Opt. Soc. Am. B* **1992**, *9*, 1100.
- [73] Z. Xie, F. Wang, C. Y. Liu, *Adv. Mater.* **2012**, *24*, 1716.
- [74] S. Kivisto, T. Hakulinen, A. Kaskela, B. Aitchison, D. P. Brown, A. G. Nasibulin, E. I. Kauppinen, A. Harkonen, O. G. Okhotnikov, *Opt. Express* **2009**, *17*, 2358.
- [75] Z. Sun, T. Hasan, A. C. Ferrari, *Phys. E* **2012**, *44*, 1082.
- [76] F. Wang, A. G. Rozhin, V. Scardaci, Z. Sun, F. Hennrich, I. H. White, W. I. Milne, A. C. Ferrari, *Nat. Nanotechnol.* **2008**, *3*, 738.
- [77] J. Wang, Y. Hernandez, M. Lotya, J. N. Coleman, W. J. Blau, *Adv. Mater.* **2009**, *21*, 2430.
- [78] G. K. Lim, Z. L. Chen, J. Clark, R. G. S. Goh, W. H. Ng, H. W. Tan, R. H. Friend, P. K. H. Ho, L. L. Chua, *Nat. Photonics* **2011**, *5*, 554.
- [79] N. J. Turro, *Modern Molecular Photochemistry*, University Science Books, Sausalito, CA, **1991**.
- [80] Y. F. Xu, Z. B. Liu, X. L. Zhang, Y. Wang, J. G. Tian, Y. Huang, Y. F. Ma, X. Y. Zhang, Y. S. Chen, *Adv. Mater.* **2009**, *21*, 1275.
- [81] D. Z. Tan, X. F. Liu, Y. Dai, G. H. Ma, M. Meunier, J. R. Qiu, *Adv. Opt. Mater.* **2015**, *3*, 836.
- [82] T. C. He, W. Wei, L. Ma, R. Chen, S. X. Wu, H. Zhang, Y. H. Yang, J. Ma, L. Huang, G. G. Gurzadyan, H. D. Sun, *Small* **2012**, *8*, 2163.
- [83] Q. L. Bao, H. Zhang, Y. Wang, Z. H. Ni, Y. L. Yan, Z. X. Shen, K. P. Loh, D. Y. Tang, *Adv. Funct. Mater.* **2009**, *19*, 3077.
- [84] I. H. Baek, H. W. Lee, S. Bae, B. H. Hong, Y. H. Ahn, D. I. Yeom, F. Rotermund, *Appl. Phys. Express* **2012**, *5*, 032701.
- [85] S. D. D. Cafiso, E. Ugoletti, A. Schmidt, V. Petrov, U. Griebner, A. Agnesi, W. B. Cho, B. H. Jung, F. Rotermund, S. Bae, B. H. Hong, G. Reali, F. Pirzio, *Opt. Lett.* **2013**, *38*, 1745.
- [86] E. Ugoletti, A. Schmidt, V. Petrov, J. W. Kim, D. I. Yeom, F. Rotermund, S. Bae, B. H. Hong, A. Agnesi, C. Fiebig, G. Erbert, X. Mateos, M. Aguilo, F. Diaz, U. Griebner, *App. Phys. Lett.* **2012**, *101*, 161112.
- [87] G. Demetriadou, H. T. Bookey, F. Biancalana, E. Abraham, Y. Wang, W. Ji, A. K. Kar, *Opt. Express* **2016**, *24*, 13033.
- [88] T. Winzer, R. Ciesielski, M. Handloser, A. Comin, A. Hartschuh, E. Malic, *Nano Lett.* **2015**, *15*, 1141.
- [89] K. J. Tielrooij, J. C. W. Song, S. A. Jensen, A. Centeno, A. Pesquera, A. Z. Elorza, M. Bonn, L. S. Levitov, F. H. L. Koppens, *Nat. Phys.* **2013**, *9*, 248.
- [90] M. Breusing, C. Ropers, T. Elsaesser, *Phys. Rev. Lett.* **2009**, *102*, 086809.
- [91] M. Breusing, S. Kuehn, T. Winzer, E. Malic, F. Milde, N. Severin, J. P. Rabe, C. Ropers, A. Knorr, T. Elsaesser, *Phys. Rev. B* **2011**, *83*, 153410.
- [92] P. A. George, J. Strait, J. Dawlaty, S. Shivaraman, M. Chandrashekar, F. Rana, M. G. Spencer, *Nano Lett.* **2008**, *8*, 4248.
- [93] C. H. Lui, K. F. Mak, J. Shan, T. F. Heinz, *Phys. Rev. Lett.* **2010**, *105*, 127404.

- [94] J. M. Dawlaty, S. Shivaraman, M. Chandrashekhara, F. Rana, M. G. Spencer, *App. Phys. Lett.* **2008**, 92, 042116.
- [95] S. Kumar, M. Anija, N. Kamaraju, K. S. Vasu, K. S. Subrahmanyam, A. K. Sood, C. N. R. Rao, *App. Phys. Lett.* **2009**, 95, 191911.
- [96] K. P. Wang, J. Wang, J. T. Fan, M. Lotya, A. O'Neill, D. Fox, Y. Y. Feng, X. Y. Zhang, B. X. Jiang, Q. Z. Zhao, H. Z. Zhang, J. N. Coleman, L. Zhang, W. J. Blau, *ACS Nano* **2013**, 7, 9260.
- [97] K. P. Wang, Y. Y. Feng, C. X. Chang, J. X. Zhan, C. W. Wang, Q. Z. Zhao, J. N. Coleman, L. Zhang, W. J. Blau, *J. Wang, Nanoscale* **2014**, 6, 10530.
- [98] Y. X. Li, N. N. Dong, S. F. Zhang, X. Y. Zhang, Y. Y. Feng, K. P. Wang, L. Zhang, J. Wang, *Laser Photonics Rev.* **2015**, 9, 427.
- [99] S. F. Zhang, N. N. Dong, N. McEvoy, M. O'Brien, S. Winters, N. C. Berner, C. Yim, Y. X. Li, X. Y. Zhang, Z. H. Chen, L. Zhang, G. S. Duesberg, J. Wang, *ACS Nano* **2015**, 9, 7142.
- [100] R. F. Wei, X. L. Tian, H. Zhang, Z. L. Hu, X. He, Z. Chen, Q. Q. Chen, J. R. Qiu, *J. Alloy. Compd.* **2016**, 684, 224.
- [101] H. Zhang, S. B. Lu, J. Zheng, J. Du, S. C. Wen, D. Y. Tang, K. P. Loh, *Opt. Express* **2014**, 22, 7249.
- [102] S. X. Wang, H. H. Yu, H. J. Zhang, A. Z. Wang, M. W. Zhao, Y. X. Chen, L. M. Mei, J. Y. Wang, *Adv. Mater.* **2014**, 26, 3538.
- [103] S. X. Wang, H. H. Yu, H. J. Zhang, *Photon. Res.* **2015**, 3, A10.
- [104] S. H. Aleithan, M. Y. Livshits, S. Khadka, J. J. Rack, M. E. Kordes, E. Stinaff, *Phys. Rev. B* **2016**, 94, 035445.
- [105] H. Y. Shi, R. S. Yan, S. Bertolazzi, J. Brivio, B. Gao, A. Kis, D. Jena, H. G. Xing, L. B. Huang, *ACS Nano* **2013**, 7, 1072.
- [106] Z. G. Nie, R. Long, L. F. Sun, C. C. Huang, J. Zhang, Q. H. Xiong, D. W. Hewak, Z. X. Shen, O. V. Prezhdo, Z. H. Loh, *ACS Nano* **2014**, 8, 10931.
- [107] Z. G. Nie, R. Long, J. S. Teguh, C. C. Huang, D. W. Hewak, E. K. L. Yeow, Z. X. Shen, O. V. Prezhdo, Z. H. Loh, *J. Phys. Chem. C* **2015**, 119, 20698.
- [108] S. Kar, Y. Su, R. R. Nair, A. K. Sood, *ACS Nano* **2015**, 9, 12004.
- [109] A. G. Čabo, J. A. Miwa, S. S. Grønberg, J. M. Riley, J. C. Johansson, C. Cacho, O. Alexander, R. T. Chapman, E. Springate, M. Grioni, *Nano Lett.* **2015**, 15, 5883.
- [110] S. B. Lu, L. L. Miao, Z. N. Guo, X. Qi, C. J. Zhao, H. Zhang, S. C. Wen, D. Y. Tang, D. Y. Fan, *Opt. Express* **2015**, 23, 11183.
- [111] K. P. Wang, B. M. Szydłowska, G. Z. Wang, X. Y. Zhang, J. J. Wang, J. J. Magan, L. Zhang, J. N. Coleman, J. Wang, W. J. Blau, *ACS Nano* **2016**, 10, 6923.
- [112] Y. H. Xu, Z. T. Wang, Z. N. Guo, H. Huang, Q. L. Xiao, H. Zhang, X. F. Yu, *Adv. Opt. Mater.* **2016**, 4, 1233.
- [113] R. Zhang, Y. X. Zhang, H. H. Yu, H. J. Zhang, R. L. Yang, B. C. Yang, Z. Y. Liu, J. Y. Wang, *Adv. Opt. Mater.* **2015**, 3, 1787.
- [114] R. J. Suess, M. M. Jadidi, T. E. Murphy, M. Mittendorff, *App. Phys. Lett.* **2015**, 107, 081103.
- [115] S. F. Ge, C. K. Li, Z. M. Zhang, C. L. Zhang, Y. D. Zhang, J. Qiu, Q. S. Wang, J. K. Liu, S. Jia, J. Feng, D. Sun, *Nano Lett.* **2015**, 15, 4650.
- [116] C. J. Zhao, H. Zhang, X. Qi, Y. Chen, Z. T. Wang, S. C. Wen, D. Y. Tang, *App. Phys. Lett.* **2012**, 101, 211106.
- [117] C. J. Zhao, Y. H. Zou, Y. Chen, Z. T. Wang, S. B. Lu, H. Zhang, S. C. Wen, D. Y. Tang, *Opt. Express* **2012**, 20, 27888.
- [118] H. Zhang, X. He, W. Lin, R. F. Wei, F. T. Zhang, X. Du, G. P. Dong, J. R. Qiu, *Opt. Express* **2015**, 23, 13376.
- [119] X. He, H. Zhang, W. Lin, R. F. Wei, J. R. Qiu, M. Zhang, B. Hu, *Sci. Rep.* **2015**, 5, 15868.
- [120] H. H. Yu, H. Zhang, Y. C. Wang, C. J. Zhao, B. L. Wang, S. C. Wen, H. J. Zhang, J. Y. Wang, *Laser Photonics Rev.* **2013**, 7, L77.
- [121] C. H. Zhu, F. Q. Wang, Y. F. Meng, X. Yuan, F. X. Xiu, H. Y. Luo, J. F. Li, X. J. Lv, L. He, Y. B. Xu, R. Zhang, *Nat. Commun.* **2017**, 8, 14111.
- [122] Z. J. Wang, H. M. Weng, Q. S. Wu, X. Dai, Z. Fang, *Phys. Rev. B* **2013**, 88, 125427.
- [123] U. Keller, *Nature* **2003**, 424, 831.
- [124] G. Bret, F. Gires, *App. Phys. Lett.* **1964**, 4, 175.
- [125] A. Miller, *J. App. Phys.* **1965**, 36, 334.
- [126] T. H. Maiman, *Nature* **1960**, 187, 493.
- [127] U. Keller, K. J. Weingarten, F. X. Kartner, D. Kopf, B. Braun, I. D. Jung, R. Fluck, C. Honninger, N. Matuschek, J. A. der Au, *IEEE J. Sel. Top. Quant.* **1996**, 2, 435.
- [128] Z. K. Liu, H. R. Mu, S. Xiao, R. B. Wang, Z. T. Wang, W. W. Wang, Y. J. Wang, X. X. Zhu, K. Y. Lu, H. Zhang, S. T. Lee, Q. L. Bao, W. L. Ma, *Adv. Mater.* **2016**, 28, 3535.
- [129] Q. B. Guo, M. X. Ji, Y. H. Yao, M. Liu, Z. C. Luo, S. A. Zhang, X. F. Liu, J. R. Qiu, *Nanoscale* **2016**, 8, 18277.
- [130] Z. P. Sun, T. Hasan, F. Torrisi, D. Popa, G. Privitera, F. Q. Wang, F. Bonaccorso, D. M. Basko, A. C. Ferrari, *ACS Nano* **2010**, 4, 803.
- [131] M. Zhang, E. J. R. Kelleher, F. Torrisi, Z. Sun, T. Hasan, D. Popa, F. Wang, A. C. Ferrari, S. V. Popov, J. R. Taylor, *Opt. Express* **2012**, 20, 25077.
- [132] G. W. Zhu, X. S. Zhu, F. Q. Wang, S. Xu, Y. Li, X. L. Guo, K. Balakrishnan, R. A. Norwood, N. Peyghambarian, *IEEE Photonics. Tech. L.* **2016**, 28, 7.
- [133] M. E. Fermann, I. Hartl, *IEEE J. Sel. Top. Quant.* **2009**, 15, 191.
- [134] A. Martinez, S. Yamashita, *App. Phys. Lett.* **2012**, 101, 041118.
- [135] W. D. Tan, C. Y. Su, R. J. Knize, G. Q. Xie, L. J. Li, D. Y. Tang, *App. Phys. Lett.* **2010**, 96, 031106.
- [136] J. L. Xu, X. L. Li, Y. Z. Wu, X. P. Hao, J. L. He, K. J. Yang, *Opt. Lett.* **2011**, 36, 1948.
- [137] J. M. Serres, P. Loiko, X. Mateos, H. H. Yu, H. J. Zhang, J. H. Liu, K. Yumashev, U. Griebner, V. Petrov, M. Aguiló, F. Diaz, *Opt. Quantum Electron.* **2016**, 48, 7.
- [138] H. H. Yu, X. F. Chen, X. B. Hu, S. D. Zhuang, Z. P. Wang, X. G. Xu, J. Y. Wang, H. J. Zhang, M. H. Jiang, *Appl. Phys. Express* **2011**, 4, 022704.
- [139] J. Ma, H. T. Huang, K. J. Ning, X. D. Xu, G. Q. Xie, L. J. Qian, K. P. Loh, D. Y. Tang, *Opt. Lett.* **2016**, 41, 890.
- [140] F. Canbaz, N. Kakenov, C. Kocabas, U. Demirbas, A. Sennaroglu, *Opt. Lett.* **2015**, 40, 4110.
- [141] M. N. Cizmeciyan, J. W. Kim, S. Bae, B. H. Hong, F. Rotermund, A. Sennaroglu, *Opt. Lett.* **2013**, 38, 341.
- [142] N. Tolstik, A. Pospischil, E. Sorokin, I. T. Sorokina, *Opt. Express* **2014**, 22, 7284.
- [143] N. Tolstik, E. Sorokin, I. T. Sorokina, *Opt. Express* **2014**, 22, 5564.
- [144] S. Husaini, R. G. Bedford, *App. Phys. Lett.* **2014**, 104, 161107.
- [145] C. A. Zaugg, Z. Sun, V. J. Wittwer, D. Popa, S. Milana, T. S. Kulmala, R. S. Sundaram, M. Mangold, O. D. Sieber, M. Golling, Y. Lee, J. H. Ahn, A. C. Ferrari, U. Keller, *Opt. Express* **2013**, 21, 31548.
- [146] U. Keller, A. C. Tropper, *Phys. Rep.* **2006**, 429, 67.
- [147] Y. W. Song, S. Y. Jang, W. S. Han, M. K. Bae, *App. Phys. Lett.* **2010**, 96, 051122.
- [148] Z. P. Sun, D. Popa, T. Hasan, F. Torrisi, F. Q. Wang, E. J. R. Kelleher, J. C. Travers, V. Nicolosi, A. C. Ferrari, *Nano Res.* **2010**, 3, 653.
- [149] Z. Q. Luo, M. Zhou, J. Weng, G. M. Huang, H. Y. Xu, C. C. Ye, Z. P. Cai, *Opt. Lett.* **2010**, 35, 3709.
- [150] Q. L. Bao, H. Zhang, Z. H. Ni, Y. Wang, L. Polavarapu, Z. X. Shen, Q. H. Xu, D. Y. Tang, K. P. Loh, *Nano Res.* **2011**, 4, 297.
- [151] W. B. Cho, J. W. Kim, H. W. Lee, S. Bae, B. H. Hong, S. Y. Choi, I. H. Baek, K. Kim, D. I. Yeom, F. Rotermund, *Opt. Lett.* **2011**, 36, 4089.
- [152] C. Wei, X. S. Zhu, F. Wang, Y. Xu, K. Balakrishnan, F. Song, R. A. Norwood, N. Peyghambarian, *Opt. Lett.* **2013**, 38, 3233.
- [153] B. Fu, Y. Hua, X. S. Xiao, H. W. Zhu, Z. P. Sun, C. X. Yang, *IEEE J. Sel. Top. Quant.* **2014**, 20, 1100705.
- [154] F. Torrisi, D. Popa, S. Milana, Z. Jiang, T. Hasan, E. Lidorskis, A. C. Ferrari, *Adv. Opt. Mater.* **2016**, 4, 1088.

- [155] Y. L. Zhong, Z. P. Cai, D. D. Wu, Y. J. Cheng, J. Peng, J. Weng, Z. Q. Luo, B. Xu, H. Y. Xu, *IEEE Photonics. Tech. L.* **2016**, 28, 1755.
- [156] J. Q. Zhao, Z. J. Zheng, D. Q. Ouyang, M. Q. Liu, X. K. Ren, S. C. Ruan, W. X. Xie, *IEEE J. Sel. Top. Quant.* **2017**, 23, 0900207.
- [157] L. Gao, T. Zhu, Y. J. Li, W. Huang, M. Liu, *IEEE Photonics. Tech. L.* **2016**, 28, 1245.
- [158] R. I. Woodward, E. J. R. Kelleher, R. C. T. Howe, G. Hu, F. Torrisi, T. Hasan, S. V. Popov, J. R. Taylor, *Opt. Express* **2014**, 22, 31113.
- [159] Z. Q. Luo, Y. Z. Huang, M. Zhong, Y. Y. Li, J. Y. Wu, B. Xu, H. Y. Xu, Z. P. Cai, J. Peng, J. Weng, *J. Lightwave. Technol.* **2014**, 32, 4077.
- [160] H. Liu, A. P. Luo, F. Z. Wang, R. Tang, M. Liu, Z. C. Luo, W. C. Xu, C. J. Zhao, H. Zhang, *Opt. Lett.* **2014**, 39, 4591.
- [161] M. Zhang, R. C. T. Howe, R. I. Woodward, E. J. R. Kelleher, F. Torrisi, G. H. Hu, S. V. Popov, J. R. Taylor, T. Hasan, *Nano Res.* **2015**, 8, 1522.
- [162] M. Liu, X. W. Zheng, Y. L. Qi, H. Liu, A. P. Luo, Z. C. Luo, W. C. Xu, C. J. Zhao, H. Zhang, *Opt. Express* **2014**, 22, 22841.
- [163] R. I. Woodward, R. C. T. Howe, T. H. Runcorn, G. Hu, F. Torrisi, E. J. R. Kelleher, T. Hasan, *Opt. Express* **2015**, 23, 20051.
- [164] J. Koo, J. Park, J. Lee, Y. M. Jhon, J. H. Lee, *Opt. Express* **2016**, 24, 10575.
- [165] Z. Q. Luo, D. D. Wu, B. Xu, H. Y. Xu, Z. P. Cai, J. Peng, J. Weng, S. Xu, C. H. Zhu, F. Q. Wang, Z. P. Sun, H. Zhang, *Nanoscale* **2016**, 8, 1066.
- [166] J. Du, Q. K. Wang, G. B. Jiang, C. W. Xu, C. J. Zhao, Y. J. Xiang, Y. Chen, S. C. Wen, H. Zhang, *Sci. Rep.* **2014**, 4, 6346.
- [167] T. Lin, H. Sun, X. Wang, D. Mao, Y. G. Wang, L. Li, L. N. Duan, *Laser Phys.* **2015**, 25, 125805.
- [168] K. Wu, X. Y. Zhang, J. Wang, J. P. Chen, *Opt. Lett.* **2015**, 40, 1374.
- [169] W. S. Li, J. Peng, Y. L. Zhong, D. D. Wu, H. Y. Lin, Y. J. Cheng, Z. Q. Luo, J. Weng, H. Y. Xu, Z. P. Cai, *Opt. Mater. Express* **2016**, 6, 2031.
- [170] R. F. Wei, H. Zhang, X. L. Tian, T. Qiao, Z. L. Hu, Z. Chen, X. He, Y. Z. Yu, J. R. Qiu, *Nanoscale* **2016**, 8, 7704.
- [171] H. Ahmad, S. N. Aitid, N. A. Hassan, M. F. Ismail, Z. C. Tiu, *Opt. Eng.* **2016**, 55, 076115.
- [172] D. Mao, B. B. Du, D. X. Yang, S. L. Zhang, Y. D. Wang, W. D. Zhang, X. Y. She, H. C. Cheng, H. B. Zeng, J. L. Zhao, *Small* **2016**, 12, 1489.
- [173] P. G. Yan, A. J. Liu, Y. S. Chen, H. Chen, S. C. Ruan, C. Y. Guo, S. F. Chen, I. L. Li, H. P. Yang, J. G. Hu, G. Z. Cao, *Opt. Mater. Express* **2015**, 5, 479.
- [174] D. Mao, Y. D. Wang, C. J. Ma, L. Han, B. Q. Jiang, X. T. Gan, S. J. Hua, W. D. Zhang, T. Mei, J. L. Zhao, *Sci. Rep.* **2015**, 5, 7965.
- [175] J. Lee, J. Park, J. Koo, Y. M. Jhon, J. H. Lee, *J. Optics* **2016**, 18, 035502.
- [176] P. G. Yan, A. J. Liu, Y. S. Chen, J. Z. Wang, S. C. Ruan, H. Chen, J. F. Ding, *Sci. Rep.* **2015**, 5, 12587.
- [177] B. H. Chen, X. Y. Zhang, K. Wu, H. Wang, J. Wang, J. P. Chen, *Opt. Express* **2015**, 23, 26723.
- [178] D. Mao, X. Y. She, B. B. Du, D. X. Yang, W. D. Zhang, K. Song, X. Q. Cui, B. Q. Jiang, T. Peng, J. L. Zhao, *Sci. Rep.* **2016**, 6, 23583.
- [179] J. Koo, Y. I. Jhon, J. Park, J. Lee, Y. M. Jhon, J. H. Lee, *Adv. Funct. Mater.* **2016**, 26, 7454.
- [180] Y. Chen, G. B. Jiang, S. Q. Chen, Z. N. Guo, X. F. Yu, C. J. Zhao, H. Zhang, Q. L. Bao, S. C. Wen, D. Y. Tang, D. Y. Fan, *Opt. Express* **2015**, 23, 12823.
- [181] H. R. Mu, S. H. Lin, Z. C. Wang, S. Xiao, P. F. Li, Y. Chen, H. Zhang, H. F. Bao, S. P. Lau, C. X. Pan, D. Y. Fan, Q. L. Bao, *Adv. Opt. Mater.* **2015**, 3, 1447.
- [182] J. Sotor, G. Sobon, W. Macherzynski, P. Paletko, K. M. Abramski, *App. Phys. Lett.* **2015**, 107, 051108.
- [183] D. D. Wu, Z. P. Cai, Y. L. Zhong, J. Peng, Y. J. Cheng, J. Weng, Z. Q. Luo, H. Y. Xu, *IEEE J. Sel. Top. Quant.* **2017**, 23, 0900106.
- [184] Y. H. Xu, Z. T. Wang, Z. N. Guo, H. Huang, Q. L. Xiao, H. Zhang, X. F. Yu, *Adv. Opt. Mater.* **2016**, 4, 1223.
- [185] J. Sotor, G. Sobon, M. Kowalczyk, W. Macherzynski, P. Paletko, K. M. Abramski, *Opt. Lett.* **2015**, 40, 3885.
- [186] D. Lee, K. Park, P. C. Debnath, I. Kim, Y. W. Song, *Nanotechnology* **2016**, 27, 365203.
- [187] L. C. Kong, Z. P. Qin, G. Q. Xie, Z. N. Guo, H. Zhang, P. Yuan, L. J. Qian, *Laser Phys. Lett.* **2016**, 13, 045801.
- [188] Z. N. Guo, H. Zhang, S. B. Lu, Z. T. Wang, S. Y. Tang, J. D. Shao, Z. B. Sun, H. H. Xie, H. Y. Wang, X. F. Yu, P. K. Chu, *Adv. Funct. Mater.* **2015**, 25, 6996.
- [189] Z. P. Qin, G. Q. Xie, H. Zhang, C. J. Zhao, P. Yuan, S. C. Wen, L. J. Qian, *Opt. Express* **2015**, 23, 24713.
- [190] Z. C. Luo, M. Liu, Z. N. Guo, X. F. Jiang, A. P. Luo, C. J. Zhao, X. F. Yu, W. C. Xu, H. Zhang, *Opt. Express* **2015**, 23, 20030.
- [191] B. T. Zhang, F. Lou, R. W. Zhao, J. L. He, J. Li, X. C. Su, J. Ning, K. J. Yang, *Opt. Lett.* **2015**, 40, 3691.
- [192] Y. X. Xie, L. C. Kong, Z. P. Qin, G. Q. Xie, J. Zhang, *Opt. Engin.* **2006**, 55, 081307.
- [193] J. F. Li, H. Y. Luo, B. Zhai, R. G. Lu, Z. N. Guo, H. Zhang, Y. Liu, *Sci. Rep.* **2016**, 6, 30361.
- [194] Z. P. Qin, G. Q. Xie, C. J. Zhao, S. C. Wen, P. Yuan, L. J. Qian, *Opt. Lett.* **2016**, 41, 56.
- [195] M. Jung, J. Lee, J. Koo, J. Park, Y.-W. Song, K. Lee, S. Lee, J. H. Lee, *Opt. Express* **2014**, 22, 7865.
- [196] J. Sotor, G. Sobon, K. M. Abramski, *Opt. Express* **2014**, 22, 13244.
- [197] K. Yin, B. Zhang, L. Li, T. Jiang, X. F. Zhou, J. Hou, *Photon. Res.* **2015**, 3, 72.
- [198] Z. C. Luo, M. Liu, H. Liu, X. W. Zheng, A. P. Luo, C. J. Zhao, H. Zhang, S. C. Wen, W. C. Xu, *Opt. Lett.* **2013**, 38, 5212.
- [199] Y. Chen, C. J. Zhao, H. H. Huang, S. Q. Chen, P. H. Tang, Z. T. Wang, S. B. Lu, H. Zhang, S. C. Wen, D. Y. Tang, *J. Lightwave. Technol.* **2013**, 31, 2857.
- [200] H. Liu, X. W. Zheng, M. Liu, N. Zhao, A. P. Luo, Z. C. Luo, W. C. Xu, H. Zhang, C. J. Zhao, S. C. Wen, *Opt. Express* **2014**, 22, 6868.
- [201] L. P. Sun, Z. Q. Lin, J. Peng, J. Weng, Y. Z. Huang, Z. Q. Luo, *Sci. Rep.* **2014**, 4, 4794.
- [202] P. H. Tang, X. Q. Zhang, C. J. Zhao, Y. Wang, H. Zhang, D. Y. Shen, S. C. Wen, D. Y. Tang, D. Y. Fan, *IEEE Photonics J.* **2013**, 5, 1500707.
- [203] P. X. Li, G. J. Zhang, H. Zhang, C. J. Zhao, J. J. Chi, Z. Q. Zhao, C. Yang, H. W. Hu, Y. F. Yao, *IEEE Photonics. Tech. L.* **2014**, 26, 1912.
- [204] S. Q. Chen, Y. Chen, M. Wu, Y. Li, C. J. Zhao, S. C. Wen, *IEEE Photonics. Tech. L.* **2014**, 26, 987.
- [205] J. Lee, J. Koo, Y. M. Jhon, J. H. Lee, *Opt. Express* **2014**, 22, 6165.
- [206] P. G. Yan, R. Y. Lin, H. Chen, H. Zhang, A. J. Liu, H. P. Yang, S. C. Ruan, *IEEE Photonics. Tech. L.* **2015**, 27, 264.
- [207] J. F. Li, H. Y. Luo, L. L. Wang, C. J. Zhao, H. Zhang, H. P. Li, Y. Liu, *Opt. Lett.* **2015**, 40, 3659.
- [208] J. Sotor, G. Sobon, W. Macherzynski, P. Paletko, K. Grodecki, K. M. Abramski, *Opt. Mater. Express* **2014**, 4, 1.
- [209] J. Sotor, G. Sobon, K. Grodecki, K. M. Abramski, *App. Phys. Lett.* **2014**, 104, 251112.
- [210] J. Tarka, J. Boguslawski, R. Zybala, M. Kowalczyk, G. Sobon, J. Sotor, *Proc. SPIE* **2016**, 9728, 972820.
- [211] M. Kowalczyk, J. Boguslawski, R. Zybala, K. Mars, A. Mikula, G. Sobon, J. Sotor, *Opt. Mater. Express* **2016**, 6, 2273.
- [212] K. S. Novoselov, A. Mishchenko, A. Carvalho, A. H. C. Neto, *Science* **2016**, 353, 461.
- [213] Y. Q. Jiang, L. L. Miao, G. B. Jiang, Y. Chen, X. Qi, X. F. Jiang, H. Zhang, S. C. Wen, *Sci. Rep.* **2015**, 5, 16372.

- [214] H. R. Mu, Z. T. Wang, J. Yuan, S. Xiao, C. Y. Chen, Y. Chen, Y. Chen, J. C. Song, Y. S. Wang, Y. Z. Xue, H. Zhang, Q. L. Bao, *ACS Photonics* **2015**, 2, 832.
- [215] Z. T. Wang, H. R. Mu, J. Yuan, C. J. Zhao, Q. L. Bao, H. Zhang, *IEEE J. Sel. Top. Quant.* **2017**, 23, 8800105.
- [216] P. Chakraborty, *J. Mater. Sci.* **1998**, 33, 2235.
- [217] G. Banfi, V. Degiorgio, D. Ricard, *Adv. Phys.* **1998**, 47, 447.
- [218] Y. Wang, *Acc. Chem. Res.* **1991**, 24, 133.
- [219] S. Q. Chen, L. L. Miao, X. Chen, Y. Chen, C. J. Zhao, S. Datta, Y. Li, Q. L. Bao, H. Zhang, Y. Liu, S. C. Wen, D. Y. Fan, *Adv. Opt. Mater.* **2015**, 3, 1769.
- [220] S. L. Yu, X. Q. Wu, K. R. Chen, B. G. Chen, X. Guo, D. X. Dai, L. M. Tong, W. T. Liu, Y. R. Shen, *Optica* **2016**, 3, 541.
- [221] A. K. Geim, I. V. Grigorieva, *Nature* **2013**, 499, 419.
- [222] U. Guler, A. Boltasseva, V. M. Shalae, *Science* **2014**, 344, 263.
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